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SMASH: Physics-guided Reconstruction of Collisions from Videos

SMASH：物理引导的视频碰撞重建

Figure 1: Starting from an input video of a collision sequence behind a curtain (top), SMASH reconstructs an accurate physically valid collision (bottom) using laws of rigid body physics for regularization.

图 1：以幕布后方发生碰撞的视频序列为输入（上），SMASH 利用刚体物理定律进行正则化，重建准确且符合物理规律的碰撞过程（下）。

Note the reconstructed spin (i.e., angular velocity) of the objects.

请注意重建出的物体旋转，即角速度。

Abstract
摘要

Collision sequences are commonly used in games and entertainment to add drama and excitement.

碰撞序列通常用于游戏和娱乐中，以增加戏剧性和刺激性。

Authoring even two body collisions in the real world can be difficult, as one has to get timing and the object trajectories to be correctly synchronized.

即便只涉及两个物体，要在真实场景中安排好一次碰撞也很困难，因为必须准确协调时间和两者的运动轨迹。

After tedious trial-and-error iterations, when objects can actually be made to collide, then they are difficult to capture in 3D.

即使经过反复试错，终于让物体发生了碰撞，要获取这一过程的三维信息仍然很困难。

In contrast, synthetically generating plausible collisions is difficult as it requires adjusting different collision parameters (e.g., object mass ratio, coefficient of restitution, etc.) and appropriate initial parameters.

同样，通过仿真生成合理的碰撞也很困难，因为需要调节碰撞参数（例如物体质量比、恢复系数等）及合适的初始参数。

We present SMASH to directly read off appropriate collision parameters directly from raw input video recordings.

我们提出 SMASH，直接从原始输入视频中估计合适的**碰撞参数**。

Technically we enable this by utilizing laws of rigid body collision to regularize the problem of lifting 2D trajectories to a physically valid 3D reconstruction of the collision.

技术上，我们利用**刚体碰撞定律**进行正则化，将二维轨迹提升为符合物理规律的三维碰撞重建。

The reconstructed sequences can then be modified and combined to easily author novel and plausible collisions.

然后可以修改和组合重建的序列，以轻松创作新颖且合理的碰撞。

We evaluate our system on a range of synthetic scenes and demonstrate the effectiveness of our method by accurately reconstructing several complex real world collision events.

我们在多种合成场景中评估系统，并通过准确重建多个复杂的真实碰撞事件，验证方法的有效性。

Keywords: 3D reconstruction, rigid body collision, optimization

关键词：3D重建，刚体碰撞，优化

Concepts: • Computing methodologies → Computer graphics; Physical simulation; Simulation evaluation; Motion processing;

概念：•计算方法→计算机图形学；物理模拟；模拟评估；运动处理；

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The definitive version was published at SIGGRAPH Asia 2016, http://dx.doi.org/10.1145/2980179.2982421.

最终版本已在 SIGGRAPH Asia 2016 上发布，http://dx.doi.org/10.1145/2980179.2982421。

1 Introduction

1 简介

Collisions capture suspense, build anticipation, and pack drama.

碰撞捕捉悬念，引发期待，并充满戏剧性。

Naturally, they remain an integral part of movies, games, and entertainment.

当然，它们仍然是电影、游戏和娱乐中不可或缺的一部分。

Creating a good real-world collision sequence involving multiple objects, however, is difficult.

然而，创建涉及多个对象的良好现实世界碰撞序列是很困难的。

While the act of smashing two objects into each other so that they collide in a certain way is already non-trivial, the setup quickly becomes unmanageable when additional colliding objects are to be collided, or adjustments are required to the object trajectories.

仅让两个物体按预期方式相撞就已不易；若再加入其他物体或调整轨迹，实验设置会很快变得难以控制。

Such changes can easily require many further iterations and recordings, and become a tedious trial-and-error process.

此类更改很容易需要许多进一步的迭代和记录，并成为一个乏味的试错过程。

Moreover, trying out multiple collision iterations with expensive or fragile objects may not even be a realistic option.

此外，尝试与昂贵或脆弱的物体进行多次碰撞迭代甚至可能不是一个现实的选择。

Accurately capturing real-world collision sequences poses further challenges.

准确捕捉现实世界的碰撞序列提出了进一步的挑战。

On one hand, such sequences necessitate high to very high framerate capture, thus making state-of-the-art methods like Kinect Fusion, etc., unsuitable candidates for 3D acquisition.

一方面，此类序列需要高到非常高的帧速率捕获，因此使得 Kinect Fusion 等最先进的方法不适合 3D 采集。

On the other hand, even high-framerate video data only provides partial 2D information, both in space and in time (see Figure 2).

另一方面，即使是高帧率视频数据也只能提供空间和时间上的部分 2D 信息（见图 2）。

A fundamental problem arises due to unavoidable occlusions near collision instances, which prevents direct observation of the actual collision processes in any acquisition setup.

碰撞发生前后不可避免的遮挡构成了一项根本困难：无论采用何种采集配置，都难以直接观测碰撞过程本身。

While the physics of object collisions is a challenging problem in itself, well-developed high-level models exist to reduce its complexity.

虽然物体碰撞的物理学本身就是一个具有挑战性的问题，但成熟的高级模型可以降低其复杂性。

One widely used assumption is that of infinite object stiffness, i.e., ideally rigid motions.

一种广泛使用的假设是无限物体刚度，即理想的刚性运动。

Whereas such rigid body simulations are widely used in games and movies, the task of setting up a collision with the right initial conditions remains tough: many parameters, such as velocities, mass ratios, and coefficient of restitution, have to be correctly guessed and adjusted.

尽管此类刚体模拟在游戏和电影中广泛使用，但在正确的初始条件下设置碰撞的任务仍然很艰巨：必须正确猜测和调整许多参数，例如速度、质量比和恢复系数。

Given the nonlinear nature of the underlying physics, such a rigging up of a desirable sequence is problematic and typically requires extensive prior experience.

考虑到基础物理的非线性性质，这种理想序列的装配是有问题的，并且通常需要大量的先前经验。

Further, physical parameters (e.g., coefficient of

此外，对于特定的一对物体，其物理参数（例如

Figure 2: Accurate acquisition of collision sequences is difficult as objects move fast and are typically occluded around collision time: a video captured by a smartphone (left) showing motion blur, and an RGBD scan using Kinect (right) that is noisy and partial.

图 2：准确采集碰撞序列很困难，因为物体移动速度很快，并且通常在碰撞时间附近被遮挡：智能手机（左）拍摄的视频显示运动模糊，而使用 Kinect（右）进行的 RGBD 扫描则有噪声且不完整。

restitution) may not be readily available for the object pair at hand.

恢复系数）也往往难以直接获得。

Finally, such an approach only enables forward simulations.

最后，这种方法只能实现正向模拟。

In this work, we propose to marry the benefits of the above setups.

在这项工作中，我们建议结合上述设置的优点。

The user records collisions between pairs of objects simply using a high framerate video (a smartphone in our setup).

用户只需使用高帧率视频（我们设置中的智能手机）即可记录成对物体之间的碰撞。

The video data, however, lacks depth information and is noisy.

然而，视频数据缺乏深度信息并且有噪声。

We first automatically extract a set of candidate 3D positions from the video, and semi-automatically initialize a sparse set of orientations.

我们首先从视频中自动提取一组候选 3D 位置，并半自动初始化一组稀疏姿态。

We then formulate an optimization using regularization derived from conservation laws of rigid body collisions to reconstruct space-time trajectories of the participating objects.

然后，我们使用从刚体碰撞守恒定律导出的正则化来制定优化，以重建参与对象的时空轨迹。

This step utilizes the original 3D models of the colliding objects, assumed to be captured beforehand.

此步骤利用碰撞对象的原始 3D 模型，假设预先捕获。

As output, we can directly read off physical collision parameters from the reconstruction that can readily be incorporated into existing physics engines to recreate and reauthor modifications of the recorded collisions.

作为输出，我们可以直接从重建中读取物理碰撞参数，这些参数可以很容易地合并到现有的物理引擎中，以重新创建和重新创作对记录的碰撞的修改。

For example, in Figure 1, we show the replay of the reconstructed collision sequence happening behind the curtain.

例如，在图 1 中，我们展示了幕后发生的重建碰撞序列的重播。

Note that we obtain high-quality reconstruction results even when the most relevant section (in time) of the original collision remains fully occluded in the input video.

请注意，即使原始碰撞的最相关部分（在时间上）在输入视频中仍然完全被遮挡，我们也可以获得高质量的重建结果。

We evaluate our method both on a range of synthetic evaluation setups and complex real-world collision examples, and generate new collision sequences using an easy authoring workflow.

我们在一系列综合评估设置和复杂的现实世界碰撞示例上评估我们的方法，并使用简单的创作工作流程生成新的碰撞序列。

In summary, we propose an algorithm to reconstruct physically valid collisions from an input video, and perform collision analysis without access to exact object geometry.

总之，我们提出了一种算法，可以从输入视频中重建物理上有效的碰撞，并在无需访问精确的对象几何形状的情况下执行碰撞分析。

The estimated collision parameters can then be used to author new collisions sequences.

然后可以使用估计的碰撞参数来创作新的碰撞序列。

2 Related work

2 相关工作

Rigid body simulation has a long and successful history within Computer Graphics.

刚体模拟在计算机图形学领域有着悠久而成功的历史。

Starting with the early works in animation [Armstrong and Green 1985; Baraff 1990], they are now widely used in all forms of computer animation.

从早期的动画作品开始[阿姆斯特朗和格林 1985； Baraff 1990]，它们现在广泛应用于各种形式的计算机动画中。

A good overview can be found in the book by Eberly [2010].

可以在 Eberly [2010] 的书中找到很好的概述。

While the forward simulation problem is far from trivial, it is well studied.

虽然正向仿真问题并不简单，但已有较为充分的研究。

We are targeting an inverse problem, and in the following we restrict discussion to the relevant works.

我们针对的是逆问题，下面我们将讨论限制在相关工作中。

The first methods to edit or modify simulations proposed spacetime methods to compute optimal motions in constrained systems [Witkin and Kass 1988; Liu et al. 1994].

第一种编辑或修改模拟的方法提出了计算约束系统中最佳运动的时空方法[Witkin and Kass 1988；刘等人。 1994]。

Variants with neural networks [Grzeszczuk et al. 1998] and genetic algorithms were proposed [Tang et al. 1995] to reduce the complexity or synthesize directable motions.

随后出现了结合神经网络 [Grzeszczuk et al. 1998] 和遗传算法 [Tang et al. 1995] 的变体，以降低复杂度或合成可控的运动。

A randomized approach to re-construct motions for target configurations of rigid bodies was proposed by Chenney et al. [2000], while an interactive variant was developed by Popovi´c et al. [2000].

Chenney 等人提出了一种随机方法来重建刚体目标配置的运动。 [2000]，而 Popovi´c 等人开发了一种交互式变体。 [2000]。

While these algorithms offer varying levels of control and parameter estimation for rigid bodies, they focus on virtual situations in a simulator.

虽然这些算法为刚体提供了不同级别的控制和参数估计，但它们侧重于模拟器中的虚拟情况。

In contrast, we propose a method that works robustly with only sparse, unreliable data from a real world source.

相比之下，我们提出了一种仅适用于来自现实世界源的稀疏、不可靠数据的稳健方法。

Areas that would be highly interesting, but which we have currently not taken into account are deformable objects [Terzopoulos et al. 1987; Martin et al. 2011] and fracture [M¨uller et al. 2001; Su et al. 2009].

可变形物体 [Terzopoulos et al. 1987; Martin et al. 2011] 和断裂 [Müller et al. 2001; Su et al. 2009] 也很值得研究，但尚未纳入本方法。

In the following, we assume that the observed objects do not deform or change topology.

在下文中，我们假设观察到的物体不会变形或改变拓扑。

Others have used the complexity of collision events to randomize the solution space, in this way giving control over the outcome of a simulation [Twigg and James 2007].

其他人利用碰撞事件的复杂性来随机化解空间，从而控制模拟的结果 [Twigg and James 2007]。

Recently, Smith et al. [2012] investigated the even tougher case of multiple, simultaneous collisions.

最近，史密斯等人 [2012] 研究了更棘手的多次同时碰撞的情况。

While we focus on the two body case, these papers highlight the complexities of collisions, which are amplified for imperfect real-world objects.

虽然我们关注的是两个物体的情况，但这些论文强调了碰撞的复杂性，对于不完美的现实世界物体，碰撞的复杂性被放大了。

A good general survey of collision modeling is the report by Gilardi and Sharf [2002].

Gilardi 和 Sharf [2002] 的报告是对碰撞建模的一个很好的综合概述。

An interesting general direction of research investigates the retrieval of modeling equations from data by employing learning techniques [Bongard and Lipson 2007].

一个有趣的一般研究方向是通过采用学习技术从数据中检索建模方程[Bongard and Lipson 2007]。

While our method is not directly using machine learning, it represents a data-driven approach to recover complex real-world phenomena.

虽然我们的方法不直接使用机器学习，但它代表了一种数据驱动的方法来恢复复杂的现实世界现象。

In this context, we demonstrate that suitable physical models are powerful regularizers for underdetermined problems.

在这一背景下，我们表明，合适的物理模型能够为欠定问题提供有效的正则化约束。

Conotter et al. [2012] analyze the kinematics of the ballistic motion of a single object in videos in the digital forensics context.

科诺特等人 [2012]在数字取证背景下分析视频中单个物体的弹道运动的运动学。

Salzmann and Urtasun [2011] explain the change in the linear motion of objects by solving for a sparse set of forces, applying their dynamics motivated regularization in video based tracking.

Salzmann 和 Urtasun [2011] 通过求解一组稀疏的力，并在基于视频的跟踪中应用其动力学驱动的正则化来解释物体线性运动的变化。

We focus on acquiring the intrinsic physical properties of the objects to guide our motion estimates taking both linear and angular motion into account.

我们专注于获取物体的内在物理属性，以指导我们的运动估计，同时考虑线性和角运动。

Geometry acquisition has been simplified with affordable and portable scanning options.

经济实惠的便携式扫描选项简化了几何采集。

Hence, in recent years, significant efforts have gone towards capturing and parsing 3D scenes.

因此，近年来，人们在捕获和解析 3D 场景方面做出了巨大的努力。

Approaches include using classifiers on scene objects [Schlecht and Barnard 2009; Xiong and Huber 2010; Anand et al. 2013; Koppula et al. 2011; Silberman et al. 2012], interactive 3D modeling from raw RGBD scans [Shao et al. 2012], interleaving segmentation and classification [Nan et al. 2012], unsupervised algorithms to identify and consolidate scans [Kim et al. 2012; Mattausch et al. 2014], dynamic reconstruction [Mitra et al. 2007], proxy geometry based scene understanding [Lafarge and Alliez 2013; Monszpart et al. 2015], or studying the spatial layout of scenes [Gupta et al. 2010; Lee et al. 2010; Hartley et al. 2012].

方法包括在场景对象上使用分类器 [Schlecht 和 Barnard 2009；熊和胡贝尔，2010；阿南德等人。 2013年；科普拉等人。 2011；西尔伯曼等人。 2012]，根据原始 RGBD 扫描进行交互式 3D 建模 [Shao 等人。 2012]，交错分割和分类[Nan 等人。 2012]，识别和整合扫描的无监督算法[Kim 等人。 2012年；马陶什等人。 2014]，动态重建[Mitra 等人。 2007]，基于代理几何的场景理解 [Lafarge and Alliez 2013；蒙斯帕特等人。 2015]，或研究场景的空间布局[Gupta 等人。 2010；李等人。 2010；哈特利等人。 2012]。

Physical constraints have also been used for scene understanding: for example, [Jia et al. 2013; Jiang and Xiao 2013; Zheng et al. 2013; Shao* and Monszpart* et al. 2014] consider local and global physical stability to predict occluded geometry in scenes.

物理约束也已用于场景理解：例如，[Jia 等人。 2013年；江、肖，2013；郑等人。 2013年； Shao* 和 Monszpart* 等人。 2014]考虑局部和全局物理稳定性来预测场景中被遮挡的几何体。

These methods, however, primarily focus on static scenes.

然而，这些方法主要关注静态场景。

In case of dynamic scenes, even state-of-the-art RGBD based systems [Newcombe et al. 2015; Dou et al. 2015] struggle with highly dynamic motion and occlusion.

在动态场景的情况下，即使是最先进的基于 RGBD 的系统 [Newcombe 等人。 2015年；窦等人。 2015] 与高动态运动和遮挡作斗争。

Capturing 3D geometry of dynamic scenes remains very challenging.

捕捉动态场景的 3D 几何仍然非常具有挑战性。

For dynamic scenarios, inverse methods with physics priors have also been used to capture other phenomena such as liquids [Wang et al. 2009] or smokes [Gregson et al. 2014].

在动态场景中，带物理先验的反演方法也用于恢复液体 [Wang et al. 2009] 和烟雾 [Gregson et al. 2014] 等现象。

While impressive results have been demonstrated in case of template-based solutions for human faces, hair, body, etc., they focus on application specific contexts where object behavior and dynamics can be captured and learned in a training phase.

虽然在针对人脸、头发、物体等的基于模板的解决方案中已经展示了令人印象深刻的结果，但它们专注于特定于应用程序的上下文，在这些上下文中可以在训练阶段捕获和学习对象行为和动态。

In contrast, we focus on reconstructing 3D geometry of collisions directly from raw videos.

相比之下，我们专注于直接从原始视频重建碰撞的 3D 几何形状。

Note that due to the nature of the problem, collisions happen prohibitively fast for accurate capture.

请注意，由于问题的性质，碰撞发生的速度非常快，无法准确捕获。

Further, object parts are severely occluded near the actual collision.

此外，物体部分在实际碰撞附近被严重遮挡。

We demonstrate, that motion states of the objects as well as internal physical properties can be estimated by analysing the objects' pre- and post-collision trajectories away from the time of collision.

我们证明，只需分析远离碰撞瞬间的碰撞前后轨迹，便可估计物体运动状态及其内在物理属性。

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3 Formulation

3 问题建模

Our goal is to acquire pairwise relevant parameters of object collisions simply based on an input video and reconstruct a physically valid motion.

我们的目标是仅根据输入视频获取对象碰撞的成对相关参数并重建物理上有效的运动。

As output we produce a space-time recording of the collision event that can be re-used in a variety of ways: to set up new collisions that behave faithfully to the original recording, to introduce new objects, or even to author complex interactions between objects by combining multiple recordings.

作为输出，我们生成碰撞事件的时空记录，可以通过多种方式重复使用：设置忠实于原始记录的新碰撞，引入新对象，甚至通过组合多个记录来创作对象之间的复杂交互。

We first briefly review rigid body dynamics (Section 3.1) as the governing equations provide important building blocks for our optimization.

我们首先简要回顾一下刚体动力学（第 3.1 节），因为控制方程为我们的优化提供了重要的构建模块。

We then formulate the problem (Section 3.2) and identify the appropriate conservation laws to constrain solutions to the space of physically-plausible ones (Section 3.3 - 3.6).

然后，我们提出问题（第 3.2 节）并确定适当的守恒定律，以将解限制在物理上合理的空间（第 3.3 - 3.6 节）。

We solve the problem in an energy-minimization framework using the above components.

我们使用上述组件在能量最小化框架中解决该问题。

In the following, we use bold lower-case letters for 3D and 4D vectors (e.g., $\mathbf{p}$), and reserve bold upper case letters for matrices (e.g., $\mathbf{R}$), see Table 1. All our measurements are in world space units.

在下文中，我们使用粗体小写字母表示 3D 和 4D 向量（例如，$\mathbf{p}$），并保留粗体大写字母表示矩阵（例如，$\mathbf{R}$），请参见表 1。我们所有的测量值均采用世界空间单位。

3.1 Rigid Bodies

3.1 刚体

By using a rigid body model we represent an object by its motion around its center of mass $\mathbf{p}$.

通过使用刚体模型，我们通过物体绕其质心的运动来表示物体 $\mathbf{p}$。

In addition to $\mathbf{p}$, each body has an orientation, represented as a unit quaternion $\mathbf{q}$ (we denote the equivalent rotation matrix by $\mathbf{R}_\mathbf{q}$).

除了$\mathbf{p}$之外，每个物体还有一个姿态，表示为单位四元数$\mathbf{q}$（我们用$\mathbf{R}_\mathbf{q}$表示等效旋转矩阵）。

Due to rigidity, we only need to consider linear and angular velocity ($\mathbf{v}$ and $\boldsymbol{\omega}$, respectively).

由于刚性，我们只需要考虑线速度和角速度（分别为$\mathbf{v}$和$\boldsymbol{\omega}$）。

A body further has a mass and a moment of inertia (calculated for a reference orientation), denoted by m and $\mathbf{I}_0$, respectively.

物体还具有质量和转动惯量（针对参考姿态计算），分别用m和$\mathbf{I}_0$表示。

While $\mathbf{p}, \mathbf{q}, \mathbf{v}$, and $\boldsymbol{\omega}$ change over time, m and $\mathbf{I}_0$ are assumed to be constant.

虽然$\mathbf{p}, \mathbf{q}, \mathbf{v}$,和$\boldsymbol{\omega}$随时间变化，但$m$和$\mathbf{I}_0$被假定为恒定。

For notational simplicity, we keep the time dependence, e.g., $\mathbf{p}(t)$, as much as possible implicit and simply write $\mathbf{p}$.

为了符号简单，我们保留时间依赖性，例如$\mathbf{p}(t)$，尽可能隐式并简单地写成$\mathbf{p}$。

The instantaneous velocity of any point $\mathbf{x}$ on a rotating body is given by $\mathbf{v} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{p})$.

旋转体上任意点$\mathbf{x}$的瞬时速度由$\mathbf{v} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{p})$给出。

Frictionless collisions between two objects are typically modelled with a scalar impulse j acting along a collision normal $\mathbf{n}$.

两个物体之间的无摩擦碰撞通常使用沿碰撞法线$\mathbf{n}$作用的标量冲量j进行建模。

For two bodies a and b, we denote their respective variables with a superscript, e.g., $\mathbf{p}^a$ is the center of mass of object a.

对于两个物体a和b，我们用上标来表示它们各自的变量，例如$\mathbf{p}^a$是物体a的质心。

We will use c for variables related to the collision event.

我们将使用 c 作为与碰撞事件相关的变量。

The impulse changes pre-collision velocities into a set of post-collision velocities (denoted with superscript pre and post, respectively) such that the scalar relative velocity $v_r = \mathbf{v}_r \cdot \mathbf{n}$ at the point of collision satisfies v^{post}

冲量将碰撞前速度改变为一组碰撞后速度（分别用上标 pre 和 post 表示），使得碰撞点处的标量相对速度$v_r = \mathbf{v}_r \cdot \mathbf{n}$满足$v^{\text{post}}$

$\dot{r} = -c$.

$\dot{r} = -c$.

Here, c is the coefficient of restitution, which is related to the amount of energy that is transferred into a reversal of the object's motion.

这里，c是恢复系数，它与传递到物体反向运动的能量有关。

The remainder is lost for the simulation and dissipated into heat, sound, or work to deform the internal structure of the objects.

其余部分在模拟中丢失，并消散为热量、声音或使物体内部结构变形的功。

In rigid body solvers we typically use a chosen value for c to compute j , and with it the post-collision velocities.

在刚体求解器中，我们通常使用c的选定值来计算j，以及碰撞后速度。

The impulse acts anti-symmetrically to conserve momentum, both linear:

两个物体受到大小相等、方向相反的冲量，以保持线动量守恒：

$$\begin{aligned} \mathbf{v}^{\text{post,a}} &= \mathbf{v}^{\text{pre,a}} + j\mathbf{n}/m^{\text{a}} , \\ \mathbf{v}^{\text{post,b}} &= \mathbf{v}^{\text{pre,b}} - j\mathbf{n}/m^{\text{b}} , \end{aligned} \qquad (1)$$

and angular, where the instantaneous change of the angular momentum $\dot{\mathbf{k}} = \mathbf{I}\boldsymbol{\omega}$ is

角动量也需守恒，其瞬时变化 $\dot{\mathbf{k}} = \mathbf{I}\boldsymbol{\omega}$ 为

$$\begin{aligned} \mathbf{k}^{\text{post,a}} &= \mathbf{k}^{\text{pre,a}} + ((\mathbf{x}^{\text{c}} - \mathbf{p}^{\text{a}}) \times j\mathbf{n}) , \\ \mathbf{k}^{\text{post,b}} &= \mathbf{k}^{\text{pre,b}} - ((\mathbf{x}^{\text{c}} - \mathbf{p}^{\text{b}}) \times j\mathbf{n}) . \end{aligned} \qquad (2)$$

3.2 Problem Statement

3.2 问题陈述

In contrast to the typical forward simulation above, our goal is to retrieve the physical parameters directly from a real collision of two

与上面典型的正向模拟相比，我们的目标是直接从两个物体的真实碰撞中检索物理参数

inputs for each ...

每个 $\ldots$ 的输入

g gravitational acceleration global f fps video frame rate recording $\mathbf{P}$ camera projection matrix recording $\mathbf{q}$ obs $_{,}$ a $_{,}\mathbf{q}$ obs $_{,}$ b orientation inputs annotation

g 重力加速度全局 f fps 视频帧率录制 $\mathbf{P}$ 相机投影矩阵录制 $\mathbf{q}$ obs $_{,}$ a $_{,}\mathbf{q}$ obs $_{,}$ b 姿态输入标注

derived quantities for each $\ldots$ $\mathbf{p}$ a $_{,}\mathbf{p}$ b object position cont. time t $\mathbf{q}$ a $_{,}\mathbf{q}$ b object orientation cont. time t $\mathbf{v}$ a $_{,}\mathbf{v}$ b linear velocity cont. time t $\bar{\boldsymbol{\omega}}$ a $_{,}\bar{\boldsymbol{\omega}}$ b angular velocity (world) cont. time t $\mathbf{I}$ a

每个 $\ldots$ $\mathbf{p}$ a, $\mathbf{p}$ b对象位置的导出量（续）时间 t $\mathbf{q}$ a $_{,}\mathbf{q}$ b 物体姿态（续）时间 t $\mathbf{v}$ a $_{,}\mathbf{v}$ b 线速度连续。时间 t $\bar{\boldsymbol{\omega}}$ a $_{,}\bar{\boldsymbol{\omega}}$ b 角速度（世界）续。时间 t $\mathbf{I}$ a

$\mathbf{I}_{0}^{a}$ b $_{0}$ inertia tensor (local) object $\mathbf{p}$ 2d $_{,}$ a $_{,}\mathbf{p}$ 2d $_{,}$ a image-space position frame (discr. time) s 2d bounding circle diameter frame (discr. time) d a $_{,}d$ b approximate depth frame (discr. time)

$\mathbf{I}_{0}^{a}$ b $_{0}$ 惯性张量（局部）物体 $\mathbf{p}$ 2d $_{,}$ a $_{,}\mathbf{p}$ 2d $_{,}$ a 像空间位置坐标系（计算时间） s 2d 边界圆直径坐标系（计算时间） d a $_{,}d$ b 近似深度坐标系（计算时间）

unknowns (number) β_x, β_{y1} gravity rotation angles recording (2) $\mathbf{b}$ a

未知数（数字） β_x, β_{y1} 重力旋转角度记录（2） $\mathbf{b}$ a

$\mathbf{b}_{3}^{a}$ b $_{3}$ parabola translation (collision pos.) object (2 $\times$ 3) $\mathbf{q}$ c,a $_{,}\mathbf{q}$ c,b orientation at collision time object (2 $\times$ 4) $\mathbf{k}$ pre $_{,}$ a $_{,}\mathbf{k}$ pre $_{,}$ b $_{,}\mathbf{k}$ post $_{,}$ a $_{,}\mathbf{k}$ post $_{,}$ b angular momentum segment (4 $\times$ 3) b pre $_{,}$ a

$\mathbf{b}_{3}^{a}$ b $_{3}$抛物线平移（碰撞位置）对象（2 $\times$ 3） $\mathbf{q}$ c,a $_{,}\mathbf{q}$ c,b 碰撞时对象的姿态（2 $\times$ 4） $\mathbf{k}$ pre $_{,}$ a $_{,}\mathbf{k}$ pre $_{,}$ b $_{,}\mathbf{k}$ post $_{,}$ a $_{,}\mathbf{k}$后,b角动量段（4×3）b前,a

$'_1$ parabola linear coefficients (x coord.) segment (4 $\times$ 1) b pre $_{,}$ a

$'_1$ 抛物线线性系数（x 坐标）段（4 $\times$ 1） b pre $_{,}$ a

$'_2$ parabola linear coefficients (y coord.) segment (4 $\times$ 1) β pre $_{,}$ a

$'_2$ 抛物线线性系数（y 坐标）线段（4 $\times$ 1） β pre $_{,}$ a

$\underline{y_0}$ parabola rotation angles segment (4 $\times$ 1)

$\underline{y_0}$抛物线旋转角度段（4×1）

$\mathbf{x}$ c collision point (world) collision (1x3) $j\mathbf{n}$ impulse times collision normal collision (1x3) t c collision time collision (1) c coefficient of restitution collision (1) m b,a mass ratio collision (1)

$\mathbf{x}$ c 碰撞点（世界）碰撞（1x3） $j\mathbf{n}$ 冲量次数碰撞 法向碰撞（1x3） t c 碰撞 时间碰撞（1） c 恢复系数碰撞（1） m b,a 质量比碰撞（1）

Table 1: Notation table.

表 1：符号表。

objects.

对象。

This would normally involve a large amount of tedious manual work or complicated capturing setups.

这通常涉及大量繁琐的手动工作或复杂的捕获设置。

We now explain our inverse approach to retrieve the physical parameters based on observations of the objects' trajectories.

我们现在解释基于对物体轨迹的观察来检索物理参数的逆方法。

It should be pointed out here that computing absolute quantities (e.g., mass) is not possible without access to a reference measurement.

这里应该指出的是，如果没有参考测量值，就不可能计算绝对数量（例如质量）。

We will not be able to compute an absolute position of the objects on earth from a single video input, and, correspondingly, we cannot compute their absolute masses.

我们将无法从单个视频输入计算地球上物体的绝对位置，相应地，我们也无法计算它们的绝对质量。

However, we can retrieve relative quantities, i.e., the relative positions, and the relative mass of the objects.

然而，我们可以检索相对量，即物体的相对位置和相对质量。

We neglect all other external effects (e.g., aerodynamic drag, friction) at work except for a known gravitational acceleration with magnitude $g = 9.81 m/s^2$.

除了已知的大小为$g = 9.81 m/s^2$的重力加速度外，我们忽略工作中的所有其他外部效应（例如，空气动力阻力、摩擦力）。

Interestingly, the actual shape of the objects does not play a direct role.

有趣的是，物体的实际形状并不起直接作用。

Only a related quantity, the distribution of mass is important in the form of the moment of inertia.

作为一个相关量，质量的分布以转动惯量的形式很重要。

We diagonalize the moment of inertia for all objects by pre-aligning the initial configuration along their principal axes.

我们通过沿主轴预先对齐初始配置来对所有物体的转动惯量进行对角化。

Thus, we represent an object' s moment of inertia with three degrees of freedom along the diagonal of $\mathbf{I}_0$,

因此，我们沿$\mathbf{I}_0$的对角线表示具有三个自由度的物体的转动惯量，

Starting from an input video, we first automatically extract a dense set of image-space centroids with associated depth values and semi-automatically generate a sparse set of orientations.

从输入视频开始，我们首先自动提取一组具有相关深度值的密集图像空间质心，并半自动生成一组稀疏姿态。

While we formulate the problem in 3D, we will later on minimize the projected error in screen space.

当我们以 3D 形式表述问题时，我们稍后将最小化屏幕空间中的投影误差。

Our goal is to compute: a parametrization of the trajectories of the two bodies a and b , their relative mass, their pre- and post-collision velocities (linear as well as angular), the time of collision, the collision impulse, and the coefficient of restitution.

我们的目标是计算：两个物体a和b轨迹的参数化、它们的相对质量、碰撞前和碰撞后的速度（线性和角度）、碰撞时间、碰撞冲量和恢复系数。

These quantities should match the observed data from the input video as well as

这些量应尽可能符合输入视频中的观测

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possible.

数据。

We will describe our approach to incorporate these unknowns in our solution in this order.

我们将按顺序描述将这些未知数合并到我们的解决方案中的方法。

As the image data is potentially unreliable and noisy, we do not rely purely on the corresponding data-terms in our calculations, but use the physical laws as regularizers.

由于图像数据可能不可靠且有噪声，因此我们在计算中并不纯粹依赖相应的数据项，而是使用物理定律作为正则化器。

One of our central contributions is identifying and incorporating the relevant physical constraints.

我们的核心贡献之一是确定并纳入相关的物理限制。

3.3 Center of Mass Trajectories

3.3 质心轨迹

A well known fact is that the trajectory of the center of mass of an object experiencing a constant acceleration is given by a parabola.

众所周知的事实是，经历恒定加速度的物体的质心轨迹由抛物线给出。

This description becomes invalid at the time of the collision t^{c}, but it is an excellent model for the trajectories before and after, as long as effects such as aerodynamic drag are negligible.

这个描述在碰撞t^{c}时变得无效，但只要空气动力阻力等影响可以忽略不计，它对于前后的轨迹来说是一个很好的模型。

For the two objects, with pre- and post-collision trajectories, we thus parametrize and extract four 3D parabolas from the inputs.

对于具有碰撞前和碰撞后轨迹的两个物体，我们从输入中参数化并提取四个 3D 抛物线。

We parametrize each parabola in 3D space with

我们对 3D 空间中的每个抛物线进行参数化

$$\mathbf{p}(t) = \mathbf{R} \begin{pmatrix} b_1(t - t^{\mathrm{c}}) \\ -\frac{g}{2}(t - t^{\mathrm{c}})^2 + b_2(t - t^{\mathrm{c}}) \\ 0 \end{pmatrix} + \mathbf{b}_3, \qquad (5)$$

where b_1, b_2 parametrize the curve over time, the vector $\mathbf{b}_3$ determines its offset, and the rotation matrix $\mathbf{R}$ its orientation.

其中b_1, b_2参数化曲线随时间的变化，矢量$\mathbf{b}_3$确定其偏移，旋转矩阵$\mathbf{R}$确定其姿态。

We shift the curve in time by the collision time t^{c}, so that $\mathbf{p}(t^{\mathrm{c}}) = \mathbf{b}_3$.

我们将曲线随时间移动碰撞时间 t_c，从而得到 $\mathbf{p}(t_c) = \mathbf{b}_3$。

Moreover, the different parabolas can only rotate about the axis of gravity, which, without loss of generality, we assume to be the y-axis in our formulation.

此外，不同的抛物线只能绕重力轴旋转，不失一般性，我们在公式中假设重力轴为 y 轴。

We encode this premise as a global rotation with two degrees of freedom shared by all four parabolas and individual rotations around the gravity direction.

为表达这一约束，我们采用一个由四条抛物线共享、具有两个自由度的全局旋转，再为各条抛物线分别设置绕重力方向的旋转。

As a consequence, we construct a rotation matrix $\mathbf{R}$ as a sequence of Euler angle rotations.

因此，我们构造一个旋转矩阵 $\mathbf{R}$作为欧拉角旋转的序列。

We chose the proper Euler angle representation Y,X,Y, the angles of which we denote with β_{y0}, β_x , and β_{y1} , respectively.

我们选择了合适的欧拉角表示Y，X，Y，我们分别用 β_{y0}, β_x和 β_{y1}表示它们的角度。

We denote the resulting rotation matrix by $\mathbf{R}_{\beta_{y0},\beta x,\beta y1}$.

我们用 $\mathbf{R}_{\beta_{y0},\beta x,\beta y1}$表示得到的旋转矩阵。

Note that β_{y0} is different for every parabola, while the other two are shared, rotating the common gravity vector.

每条抛物线的 β_{y0} 不同，另外两个角度由所有抛物线共享，用于旋转共同的重力向量。

Further, the two parabolas for an object coincide in one point at the time of collision, allowing us to express both parabolas uniquely with a single $\mathbf{b}_3$ offset.

此外，碰撞时物体的两条抛物线在一点上重合，因此我们可以通过一个 $\mathbf{b}_3$偏移来唯一地表达两条抛物线。

In total, that leaves us with two global angles, four individual angles, two offsets, and four times b_1, b_2 as unknowns.

最终未知量包括 2 个全局角度、4 个独立角度、2 个平移向量，以及 4 组 b_1, b_2。

The geometric setup of the four parabolas is illustrated in Figure 3.

四个抛物线的几何设置如图 3 所示。

We directly estimate the linear velocity from these curves with the temporal derivative of Eq. (5).

我们用方程的时间导数直接估计这些曲线的线速度。 （5）。

The most important velocity in our setting is the velocity at collision time t^c:

我们设置中最重要的速度是碰撞时的速度 t^c：

$$\mathbf{v}(t^c) = \mathbf{R} \begin{pmatrix} b_1 & b_2 - gt^c & 0 \end{pmatrix}^T \qquad (6)$$

the angular aspects of our motion modeling.

We now describe the angular aspects of our motion modeling.

我们现在描述运动建模的角度方面。

3.4 Orientation and Angular Momentum

3.4 姿态和角动量

While the angular velocity of a rigid body can change without external forces, its angular momentum remains constant (see [Kleppner and Kolenkow 2013]).

虽然刚体的角速度可以在没有外力的情况下发生变化，但其角动量保持恒定（参见 [Kleppner 和 Kolenkow 2013]）。

Thus, for each of the four trajectories, we calculate an angular momentum $\mathbf{k}$ that best explains the orientations observed over time.

因此，对于四个轨迹中的每一个，我们计算一个角动量 $\mathbf{k}$，它最好地解释了随着时间的推移观察到的姿态。

Angular momentum and velocity relate to each other with:

角动量和速度彼此相关：

$$\boldsymbol{\omega}(t) = \mathbf{I}^{-1}\mathbf{k}(t) = \mathbf{R}_\mathbf{q}(t)\mathbf{I}_0^{-1}\mathbf{R}_\mathbf{q}^{-1}(t)\mathbf{k}(t) \qquad (7)$$

Once we know the angular momentum at time t, we can compute the corresponding angular velocity $\boldsymbol{\omega}$, and use it to integrate an

已知 t 时刻的角动量，就能计算对应角速度 $\boldsymbol{\omega}$，再通过积分

orientation forward in time.

推算后续时刻的姿态。

With $\boldsymbol{\omega}$ as the imaginary part of a quaternion, an Euler step is given by:

将$\boldsymbol{\omega}$作为四元数的虚部，欧拉步由下式给出：

$$\mathbf{q}(t+\Delta t) = \mathbf{q}(t) + \frac{\Delta t}{2} \begin{pmatrix} 0 \\ \boldsymbol{\omega} \end{pmatrix} \otimes \mathbf{q}(t) \qquad (8)$$

where $\otimes$ denotes the quaternion product, and we normalize $\mathbf{q}(t+\Delta t)$ after each step.

其中$\otimes$表示四元数乘积，我们在每一步之后对$\mathbf{q}(t+\Delta t)$进行归一化。

Similar to the shared offset of the parabolas, we introduce an orientation $\mathbf{q}'$c for each body at collision time that is shared by the two parabola segments.

与抛物线的共享偏移类似，我们在碰撞时为每个物体引入由两个抛物线段共享的姿态$\mathbf{q}'$c。

Our goal is not to use one of the input orientations directly, due to their potential unreliability.

我们的目标是不直接使用输入姿态之一，因为它们潜在的不可靠性。

Instead we find an improved orientation that, together with an angular momentum $\mathbf{k}$, explains the observed orientations as well as possible.

相反，我们找到了一种改进的姿态，它与角动量一起$\mathbf{k}$，尽可能地解释了观察到的姿态。

Starting with $\mathbf{q}'$c at collision time t'c we integrate this orientation backward or forward to time t with explicit integration.

从碰撞时间t'c处的$\mathbf{q}'$c开始，我们通过显式积分将该姿态向后或向前积分到时间t。

We perform a series of integration steps with Eq. (8) from t'c to a given time t.

我们使用等式执行一系列积分步骤。 (8) 从t'c到给定时间t。

This yields the orientation of body a or b at any instance in time based on the estimated orientation at collision time.

根据碰撞时的估计姿态，这会及时产生物体a或b在任何时刻的姿态。

We will later on use this step to calculate the estimated orientation at different times, based on our current estimate for the collision orientation and object motion.

稍后我们将根据当前对碰撞姿态和物体运动的估计，使用此步骤来计算不同时间的估计姿态。

Thus it will later on serve to guide the optimization towards the input orientations.

因此，它将稍后用于指导输入姿态的优化。

3.5 Energy Limiting

3.5 能量限制

When gravity is the only external force present, the sum of kinetic and potential energy (for height h) remains constant as:

当重力是唯一存在的外力时，动能和势能之和（对于高度 h）保持恒定，如下所示：

$$E = \frac{1}{2}m\,\mathbf{v}^T\mathbf{v} + \frac{1}{2}\boldsymbol{\omega}^T\mathbf{I}\boldsymbol{\omega} + mgh. \qquad (9)$$

Upon collision, a portion of this energy dissipates.

碰撞时，部分能量消散。

This is typically modeled with the aforementioned coefficient of restitution.

这通常是用前面提到的恢复系数来建模的。

As it is one of the central parameters that we want to retrieve, we cannot directly impose constraints conserving the energy in our system.

由于它是我们想要检索的中心参数之一，因此我们不能直接施加约束来保存系统中的能量。

However, assuming the collision is an instantaneous event, we can ensure that the post-collision energy is less or equal to the pre-collision energy.

假设碰撞瞬时发生，就可以约束碰撞后能量不大于碰撞前能量。

As position and orientation do not change at the time of collision, the potential energy remains the same.

由于碰撞时位置和姿态不变，因此势能保持不变。

Thus, our formulation focuses on the kinetic energy, and we will later on ensure that it does not increase after the collision.

因此，我们的公式重点关注动能，稍后我们将确保碰撞后动能不会增加。

3.6 Conservation of Momentum

3.6 动量守恒

Throughout the objects' trajectories, angular momentum is conserved in the absence of external torques.

在物体的整个轨迹中，在没有外部扭矩的情况下，角动量是守恒的。

Linear momentum is only conserved in the direction orthogonal to gravity.

线性动量仅在与重力正交的方向上守恒。

Instantaneous rigid body collisions fully preserve both total linear and total angular momentum of the objects involved.

瞬时刚体碰撞完全保留了所涉及物体的总线性动量和总角动量。

As such, the equations for an impulse-based collision are ideal to couple the four separate trajectories at the time of collision.

因此，基于冲量的碰撞方程非常适合在碰撞时耦合四个单独的轨迹。

We have already exploited the fact that the two parabolas for each object have to connect in the respective collision positions, additionally, the conservation laws for momentum enable the coupling of the two objects' trajectories.

前面已利用同一物体的两条抛物线必须在碰撞位置相接这一条件；动量守恒进一步将两个物体的轨迹耦合起来。

Conservation of linear momentum means that the sum of linear momenta before and after collision has to be equal.

线动量守恒是指碰撞前后的线动量之和必须相等。

Since we cannot recover the absolute mass, we set the mass of body a to 1.0, and introduce a mass ratio m_b/m_a, (we will revisit the case of a potentially infinitely heavy object in Section 5.2).

由于无法恢复绝对质量，我们将物体 a 的质量设为 1.0，并引入质量比 m_b/m_a；第 5.2 节将讨论其中一个物体质量可能视为无限大的情形。

Formulating the equations in terms of m_b/m_a yields

根据 m_b/m_a 制定方程，得出

$$\mathbf{v}^{\text{pre,a}}(t^{\text{c}}) + \mathbf{v}^{\text{pre,b}}(t^{\text{c}})m^{\text{b,a}} = \mathbf{v}^{\text{post,a}}(t^{\text{c}}) + \mathbf{v}^{\text{post,b}}(t^{\text{c}})m^{\text{b,a}}\,. \quad (10)$$

As the velocities are determined from the parabolas, this equation ensures that these curves do not create or lose linear momentum, and effectively couple the planes of the four parabolas.

速度由抛物线决定，因此该方程保证这些曲线不会凭空增加或减少线动量，并有效耦合 4 条抛物线所在的平面。

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Figure 3: Modeling two-body collisions: We extract dense position estimates from a video input and augment it with sparse semi-automatically generated orientations.

图 3：双刚体碰撞建模。先从视频提取密集的位置估计，再辅以半自动获得的稀疏姿态估计。

These observations are linked with laws of physics: the individual trajectories should be parabolas as we assume the bodies to have ballistic motion, and they are coupled at the (unknown) collision point based on conservation laws.

物理定律将观测联系起来：假设物体做抛体运动，各段轨迹应为抛物线；再根据守恒定律，在未知碰撞点处将它们耦合。

k post ,a k post ,b

k post ,a k post ,b

We similarly ensure that the angular momentum we compute does not violate conservation of angular momentum.

同样，我们确保我们计算的角动量不违反角动量守恒。

For the angular momenta, we have to consider their sum, and additionally the instantaneous angular momentum around the origin:

对于角动量，我们必须考虑它们的总和，另外还要考虑原点周围的瞬时角动量：

$$\mathbf{p}^{\text{a}} \times \mathbf{v}^{\text{pre,a}} + \mathbf{k}^{\text{pre,a}} + m^{\text{b,a}}\mathbf{p}^{\text{b}} \times \mathbf{v}^{\text{pre,b}} + \mathbf{k}^{\text{pre,b}} = \mathbf{p}^{\text{a}} \times \mathbf{v}^{\text{post,a}} + \mathbf{k}^{\text{post,a}} + m^{\text{b,a}}\mathbf{p}^{\text{b}} \times \mathbf{v}^{\text{post,b}} + \mathbf{k}^{\text{post,b}} \quad (11)$$

Note, that we use m b,a to compute I b

注意，我们使用 $m^{\mathrm{b,a}}$ 来计算 $\mathbf{I}^{\mathrm{b}}$

$\check{\mathrm{o}}$ and $\mathbf{k}^{\mathrm{b}}$ using Eq. (3).

$\check{\mathrm{o}}$和 $\mathbf{k}^{\mathrm{b}}$ 使用式（3）。

All quantities in Eq. (11) are evaluated at collision time t^{c}.

方程式中的所有数量。(11) 在碰撞时计算 t^{c}。

The two equations (10) and (11) constrain our solution space to a physically plausible one, and provide a first coupling between objects a and b.

式（10）和（11）将解限制在物理合理的范围内，并在物体 a 与 b 之间建立初步耦合。

We improve this coupling with the help of the impulse equations from Section 3.1.

我们借助 3.1 节中的冲量方程改进了这种耦合。

The equations for linear impulse (Eq. (1)) express the exchange of momentum between the two bodies relative to their mass:

线性冲量方程（方程（1））表示两个物体之间相对于其质量的动量交换：

$$\begin{aligned}\mathbf{v}^{\mathrm{post,a}}(t^{\mathrm{c}}) &= \mathbf{v}^{\mathrm{pre,a}}(t^{\mathrm{c}}) + \frac{j\mathbf{n}}{1}\,,\\ \mathbf{v}^{\mathrm{post,b}}(t^{\mathrm{c}}) &= \mathbf{v}^{\mathrm{pre,b}}(t^{\mathrm{c}}) - \frac{j\mathbf{n}}{m^{\mathrm{b,a}}}\,. \qquad (12)\end{aligned}$$

This equation turns out to be crucial for retrieving the mass ratio $m^{\mathrm{b,a}}$ during the combined optimization, which we will detail below.

下文的联合优化中，该方程对恢复质量比 $m^{\mathrm{b,a}}$ 至关重要。

The angular impulse equations allow us to enforce a shared collision point $\mathbf{x}^{\mathrm{c}}$.

角冲量方程允许我们强制执行共享碰撞点 $\mathbf{x}^{\mathrm{c}}$。

Eq. (2) connects the angular momenta of the two bodies via the relative offset of the collision point:

式（2） 通过碰撞点的相对偏移量连接两个物体的角动量：

$$\mathbf{k}^{\text{post,a}} = \mathbf{k}^{\text{pre,a}} + ((\mathbf{x}^{\text{c}} - \mathbf{b}_3^{\text{a}}) \times j\mathbf{n})$$
$$\mathbf{k}^{\text{post,b}} = \mathbf{k}^{\text{pre,b}} - ((\mathbf{x}^{\text{c}} - \mathbf{b}_3^{\text{b}}) \times j\mathbf{n}) \,. \qquad (13)$$

We complete the set of necessary equations by computing the relative velocities at the collision point to compute the coefficient of restitution.

最后，计算碰撞点处的相对速度以求得恢复系数，从而补全所需方程。

We only need to consider the velocity components along the collision normal direction:

我们只需要考虑沿碰撞法线方向的速度分量：

$$c = -\frac{\left(\hat{\mathbf{v}}^{\text{post,a}} - \hat{\mathbf{v}}^{\text{post,b}}\right) \cdot \mathbf{n}}{\left(\hat{\mathbf{v}}^{\text{pre,a}} - \hat{\mathbf{v}}^{\text{pre,b}}\right) \cdot \mathbf{n}}, \qquad (14)$$

with $\hat{\mathbf{v}} = \mathbf{v}(t^{\text{c}}) + \boldsymbol{\omega}(t^{\text{c}}) \times (\mathbf{x}^{\text{c}} - \mathbf{p}(t^{\text{c}}))$ and $\boldsymbol{\omega}(t^{\text{c}}) = \mathbf{I}^{-1}(t^{\text{c}})\,\mathbf{k}(t^{\text{c}})$.

与 $\hat{\mathbf{v}} = \mathbf{v}(t^{\text{c}}) + \boldsymbol{\omega}(t^{\text{c}}) \times (\mathbf{x}^{\text{c}} - \mathbf{p}(t^{\text{c}}))$ 和 $\boldsymbol{\omega}(t^{\text{c}}) = \mathbf{I}^{-1}(t^{\text{c}})\,\mathbf{k}(t^{\text{c}})$。

With these equations at hand we now formulate an optimization problem to compute the unknown quantities as described next.

有了这些方程，我们现在制定一个优化问题来计算未知量，如下所述。

4 Method

4 方法

Our method proceeds in three stages: First, we estimate center of mass positions based on the input video.

我们的方法分三个阶段进行：首先，我们根据输入视频估计质心位置。

Second, we initialize orientations using a differential raytracer, with user input for ambiguous situations.

第二步，利用可微光线追踪器初始化姿态；存在歧义时引入用户输入。

Third, in the main step, we take these position and orientation estimates to retrieve the physical parameters of the rigid body motion and collision.

第三步是核心步骤：根据上述位置与姿态估计，恢复刚体运动及碰撞的物理参数。

For the initialization, we use a stripped-down version of our main optimization step.

对于初始化，我们使用主要优化步骤的精简版本。

We first explain the main step of the algorithm, before detailing the two initialization stages.

在详细说明两个初始化阶段之前，我们首先解释该算法的主要步骤。

4.1 Parameter Estimation

4.1 参数估计

The quantities described in Section 3 can be divided into those related to the input data and those related to the underlying physics.

第 3 节中描述的量可以分为与输入数据相关的量和与基础物理相关的量。

In terms of unknowns, we distinguish between the parameters of the four trajectory segments of the two objects, which determine position and velocities, and the unknowns of the collision event.

在未知数方面，我们区分两个物体的四个轨迹段的参数（确定位置和速度）以及碰撞事件的未知数。

We will use a superscript $*$ to refer to the four trajectories, i.e., $* \in \{$ pre-a , pre-b , post-a , post-b $\}$ and $\diamond$ for objects, i.e., $\diamond \in \{$ a , b $\}$.

我们用上标 $*$来指代四个轨迹，即$* \in \{$pre-a,pre-b,post-a,post-b$\}$和 $\diamond$对于物体，即 $\diamond \in \{$a,b$\}$。

For each trajectory, we have the unknowns of each parabola.

对于每条轨迹，我们都有每条抛物线的未知数。

They are b_1^*

他们是 b_1^*

$\check{3}$ exist once for each body.

$\check{3}$每个物体存在一次。

Also, we have the global angles β_x, β_{y1} , and unknowns for angular momentum exist for each trajectory segment, $\mathbf{k}^*$.

此外，我们还有全局角度 β_x, β_{y1}，并且每个轨迹段都存在角动量未知数 $\mathbf{k}^*$。

Additionally, the unknowns for the collision are: orientation for both objects at collision time $\mathbf{q}$ c ,$\diamond$, location $\mathbf{x}$ c, impulse $j\mathbf{n}$, and the mass ratio m b,a, as well as the coefficient of restitution c and the time of the collision t c.

此外，碰撞未知数为：碰撞时两个物体的姿态$\mathbf{q}$c,⋄、位置$\mathbf{x}$c、冲量$j\mathbf{n}$、质量比mb、a，以及恢复系数c和碰撞时间tc。

Our optimization uses a dense set of 2D object centroids $\{\mathbf{p}$ 2d

我们的优化使用一组密集的 2D 对象质心$\{\mathbf{p}$ 2d

$i=1$ with depth values $\{d_i\}_{i=}^{N}$

$i=1$ 带深度值 $\{d_i\}_{i=}^{N}$

$i=1$ at times $\{t$ obs

$i=1$，对应观测时刻 $\{t$ obs

$i=1$, and a sparse set of orientations for both bodies $\{\mathbf{q}$ obs

$i=1$ ，以及两个物体的一组稀疏姿态 $\{\mathbf{q}$ obs

$j=1$ at times $\{t$ obs

$j=1$，对应观测时刻 $\{t$ obs

$j=1$ from the input video.

$j=1$来自输入视频。

Later, in Section 4.2, we describe how to extract this data from the video.

稍后，在 4.2 节中，我们将描述如何从视频中提取这些数据。

We use between 200 and 350 video frames with position information, and two semi-automatically annotated orientations per parabola segment, thus eight orientations in total.

我们使用 200 到 350 个带有位置信息的视频帧，以及每个抛物线段两个半自动标注的姿态，因此总共有八个姿态。

Additionally, we require a bounding box estimate for each body: $\mathbf{s}^{\diamond}$.

此外，我们需要每个物体的边界框估计：$\mathbf{s}^{\diamond}$。

Finally, we have a set of additional equations from the conservation laws, namely Equations (10-14) to tie the observations together according to the physical model.

最后，我们有一组来自守恒定律的附加方程，即方程（10-14），用于根据物理模型将观测结果联系在一起。

They represent our physics-based

它们代表了我们基于物理的

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regularization encoded as additional equations.

正则化编码为附加方程。

As the number of equations from input data and physics is significantly larger than the number of unknowns, we have an over-determined problem.

由于输入数据和物理方程的数量明显大于未知数的数量，因此我们遇到了超定问题。

As the equations are non-linear, we compute a solution using non-linear least-squares.

由于方程是非线性的，我们使用非线性最小二乘法计算解。

We summarize the unknowns in the vector θ for this least-squares optimization, and reformulate the equations to take the form

将最小二乘优化的未知量集中记为向量 θ，可将方程改写为

$$\underset{\theta}{\operatorname{argmin}} \sum_{\tau \in \mathrm{Terms}} w^{\tau} f^{\tau}(\theta) , \qquad (15)$$

where w^{τ} is weighting the corresponding energy term f^{τ}, and $\mathrm{Terms} = \{\mathrm{mom}, \mathrm{imp}, \mathrm{g}, \mathrm{ke}, \mathrm{CoR}, \mathrm{pxy}, \mathrm{pz}, \mathrm{ori}\}$.

其中 w^{τ} 是对应能量项 f^{τ} 的权重，且 $\mathrm{Terms} = \{\mathrm{mom}, \mathrm{imp}, \mathrm{g}, \mathrm{ke}, \mathrm{CoR}, \mathrm{pxy}, \mathrm{pz}, \mathrm{ori}\}$。

We now describe these individual energy terms.

下面逐一说明这些能量项。

Respecting physics.

满足物理约束。

We directly turn the physics constraints in Equations (10-14) into energy terms by defining the least squares energy to be their residual.

以式（10—14）的残差定义最小二乘能量，将物理约束直接转化为能量项。

We combine the equations for momentum conservation (10,11) into f mom:

将动量守恒方程（10、11）合并为 f mom：

$$
\begin{aligned}
f^{\text{mom}} = & \frac{1}{2}\|\mathbf{v}^{\text{pre,a}} + \mathbf{v}^{\text{pre,b}} m^{\text{b,a}} - \mathbf{v}^{\text{post,a}} - \mathbf{v}^{\text{post,b}} m^{\text{b,a}}\|^2 \qquad (16) \\
& + \frac{1}{2}\|\mathbf{p}^{\text{a}} \times \mathbf{v}^{\text{pre,a}} + \mathbf{k}^{\text{pre,a}} + m^{\text{b,a}} \mathbf{p}^{\text{b}} \times \mathbf{v}^{\text{pre,b}} + \mathbf{k}^{\text{pre,b}} - \\
& \quad \mathbf{p}^{\text{a}} \times \mathbf{v}^{\text{post,a}} - \mathbf{k}^{\text{post,a}} - m^{\text{b,a}} \mathbf{p}^{\text{b}} \times \mathbf{v}^{\text{post,b}} - \mathbf{k}^{\text{post,b}}\|^2 ,
\end{aligned}
$$

and the impulse equations (12,13) into f imp:

将冲量方程（12、13）合并为 f imp：

$$
\begin{aligned}
f^{\text{imp}} = & \frac{1}{4}\Big(\|\mathbf{v}^{\text{pre,a}} + \frac{j\mathbf{n}}{1} - \mathbf{v}^{\text{post,a}}\|^2 + \|\mathbf{v}^{\text{pre,b}} - \frac{j\mathbf{n}}{m^{\text{b,a}}} - \mathbf{v}^{\text{post,b}}\|^2 \\
& + \|\mathbf{k}^{\text{pre,a}} + ((\mathbf{x}^{\text{c}} - \mathbf{b}_3^a) \times j\mathbf{n}) - \mathbf{k}^{\text{post,a}}\|^2 \\
& + \|\mathbf{k}^{\text{pre,b}} - ((\mathbf{x}^{\text{c}} - \mathbf{b}_3^b) \times j\mathbf{n}) - \mathbf{k}^{\text{post,b}}\|^2 \Big) . \qquad (17)
\end{aligned}
$$

Both terms are evaluated at collision time t^{c}, omitted for brevity. As

Both terms are evaluated at collision time t^{c}, omitted for brevity.

这两项均在碰撞时刻 t^{c} 计算，为简洁起见未写出时间参数。

As our inputs are typically hand-held videos, we do not fully constrain gravity to act along the y-axis, but allow the optimization to slightly deviate, if necessary.

输入通常来自手持拍摄，因此不将重力方向严格固定在 y 轴，而是允许优化在必要时作小幅调整。

For this we define the energy term

为此，定义能量项

$$
f^{\text{g}} = \left(g - \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \mathbf{R}_{0,\beta_x,\beta_{y1}} \begin{pmatrix} 0 & g & 0 \end{pmatrix}^T \right)^2 . \qquad (18)
$$

that β_{y0} is zero here, as this value represents rotation around

Note that β_{y0} is zero here, as this value represents rotation around the axis of gravity.

此处 β_{y0} 为零，因为它表示绕重力轴的旋转。

We formulate the kinetic energy limiting term from Eq. (9) as:

由式（9）定义动能限制项：

$^{\text{pre}})^2$ for E post

$^{\text{pre}})^2$，当 E post 时

where pre- and post-collision kinetic energies are evaluated in terms of velocities at the time of collision ($\times \in \{$ pre , post $\}$):

碰撞前后动能由碰撞时刻的速度计算（$\times \in \{$ pre , post $\}$）：

$$E_k^{\times} = \mathbf{v}^{\times,\mathrm{a}}(t^{\mathrm{c}})^2 + \boldsymbol{\omega}^{\times,\mathrm{a}}(t^{\mathrm{c}})^T \mathbf{I}^{\mathrm{a}}(t^{\mathrm{c}})\, \boldsymbol{\omega}^{\times,\mathrm{a}}(t^{\mathrm{c}}) + m^{\mathrm{b,a}}\, \mathbf{v}^{\times,\mathrm{b}}(t^{\mathrm{c}})^2 + \boldsymbol{\omega}^{\times,\mathrm{b}}(t^{\mathrm{c}})^T \mathbf{I}^{\mathrm{b}}(t^{\mathrm{c}})\, \boldsymbol{\omega}^{\times,\mathrm{b}}(t^{\mathrm{c}})\,. \qquad (20)$$

Finally, to disallow energy creation, we directly constrain the c estimate to physical range of zero to one.

最后，为了禁止能量产生，我们直接将 c估计限制在零到一的物理范围内。

For this we include a term

为此加入如下项

c^2 for $c < 0$ $(c-1)^2$ for $c > 1$ 0 otherwise ,

c^2对于 $c < 0$$(c-1)^2$对于 $c > 1$0否则，

with c calculated using Equation (14).

c用式（14）计算。

Aligning to data. Next, we formulate the data terms. Based on Eq. (5) we add terms for position fidelity. The notation is

拟合观测数据。接下来构造数据项：根据式（5）加入位置保真项。符号略显复杂，

Figure 4: In order to automatically retrieve measurements of the object centroids, we first post-process the input video (a) to subtract the background.

图 4：为了自动检索对象质心的测量值，我们首先对输入视频 (a) 进行后处理以减去背景。

This step yields approximate, and noisy image regions for the objects (b).

此步骤生成对象 (b) 的近似噪声图像区域。

slightly complicated by the fact that we do not directly compute the difference of two 3D positions, but instead separately consider screen space and depth, each of which we will detail below.

是因为我们没有直接计算两个三维位置之差，而是分别处理图像平面和深度，下面依次说明。

For the screen space positions $\mathbf{p}$ 2d $,*$

对于屏幕空间位置 $\mathbf{p}$ 2d $,*$

i (each at time t obs

i （每次 t obs

i) that were extracted from the video, we add an energy term penalizing screen-space deviations of the corresponding parabola with

i）从视频中提取，我们添加一个能量项来惩罚相应抛物线的屏幕空间偏差

$$\mathbf{p}_i^*(t_i^{\text{obs}}) = \begin{pmatrix} b_1^*(t_i^{\text{obs}} - t^{\text{c}}) \\ -\frac{g}{2}(t_i^{\text{obs}} - t^{\text{c}})^2 + b_2^*(t_i^{\text{obs}} - t^{\text{c}}) \\ 0 \end{pmatrix} + \mathbf{b}_3^{\diamond}$$

$$f^{\text{pxy}} = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{P}\ \mathbf{p}_i^*(t_i^{\text{obs}}) - \mathbf{p}_i^{\text{2d},*}\|^2\ , \qquad (22)$$

where $\mathbf{P}$ is the camera projection matrix, $N = |\{\mathbf{p}$ 2d $|,*$

其中 $\mathbf{P}$是相机投影矩阵， $N = |\{\mathbf{p}$ 2d $|,*$

i $\}|$ and $*$ is again one of $\{$ pre-a $,$ pre-b $,$ post-a $,$ post-b $\}$ depending on the input i , and $\diamond \in \{$ a $,$ b $\}$.

i $\}|$和 $*$又是 $\{$pre-a$,$pre-b$,$post-a$,$post-b$\}$之一，取决于输入i和 $\diamond \in \{$a$,$b$\}$。

Note that $\mathbf{b}$ pre $,\diamond$

请注意，$\mathbf{b}$先于$,\diamond$

$_3$ is shared among preand post-collision parabolas of the same object.

$_3$同一物体的碰撞前和碰撞后抛物线共享。

For the depth, we compute the energy terms with

对于深度，我们计算能量项

$$f^{\mathrm{pz}} = \frac{1}{N}\sum_{i=1}^{N}\|\begin{pmatrix}0 & 0 & 1\end{pmatrix}\mathbf{p}_i^*(t_i^{\mathrm{obs}}) - d_i\|_\epsilon\,, \qquad (23)$$

where $\|\cdot\|_\epsilon$ denotes a robust Huber [1964] loss function to allow for well-supported changes in depth.

其中$\|\cdot\|_\epsilon$表示稳健的 Huber [1964] 损失函数，以允许良好支持的深度变化。

Specifically,

具体来说，

$$\|x\|_\epsilon = \begin{cases} x^2/(2\epsilon) & \text{if } x^2 \le \epsilon \\ |x| - \epsilon/2 & \text{otherwise.} \end{cases} \qquad (24)$$

The energy terms for the orientations are straight-forward to add based on Eq. (8):

姿态的能量项可以根据方程直接添加。（8）：

$$f_j^{\mathrm{ori}} = \frac{1}{M}\sum_{j=1}^{M}\|\mathbf{q}^*(t_j^{\mathrm{obs}}) - \mathbf{q}_j^{\mathrm{obs},*}\|^2\,, \qquad (25)$$

where $M = |\{\mathbf{q}_j^{\mathrm{obs},*}\}|$, and $\mathbf{q}^*(t_j^{\mathrm{obs}})$ is expressed as a series of

$^{\mathrm{obs}})$ is expressed as a series of explicit integration steps starting from $\mathbf{q}^{c,\diamond}, \diamond \in \{a, b\}$.

$^{\mathrm{obs}})$表示为从$\mathbf{q}^{c,\diamond}, \diamond \in \{a,b\}$开始的一系列显式积分步骤。

We use the analytic equations for cuboids to calculate the moment of inertia for our bodies.

我们采用长方体的解析公式计算物体的转动惯量。

We pre-align their principal moments with the coordinate axes using SVD on the scanned models during a pre-computation

在预计算期间，我们在扫描模型上使用 SVD 将它们的主矩与坐标轴预先对齐

Figure 5: The steps of our position initialization: (a) object region centroids; (b) initial labeling: pre-collision in blue, post-collision in red, and discarded points in gray; (c) re-assignment, and (d) final parabolas: object A in blue, object B in purple.

图 5：位置初始化的步骤：(a) 对象区域质心； (b) 初始标记：碰撞前为蓝色，碰撞后为红色，丢弃点为灰色； (c) 重新分配，以及 (d) 最终抛物线：对象 A 为蓝色，对象 B 为紫色。

step.

步。

We then scale it with the masses of the objects, i.e., 1.0 and m b,a.

然后我们用物体的质量来缩放它，即 1.0 和m b,a。

For a specific instance in time, $\mathbf{I}_0$ is transformed with the orientation at this time according to Equation (4).

对于一个特定的时间实例，$\mathbf{I}_0$根据方程（4）与此时的姿态进行变换。

Collision point.

碰撞点。

As our optimization has to navigate a highly non-linear energy landscape, a good initialization is crucial (see Sections 4.2 and 4.3).

目标能量高度非线性，因此良好的初始化对优化求解至关重要（见第 4.2、4.3 节）。

Additionally, we found that our optimization profits from constraining the collision point at first.

此外，我们发现我们的优化首先从限制碰撞点中受益。

We thus run our full optimization twice: first (Alg. 1, line 30) with a collision point $\mathbf{x}$ c $=(\mathbf{b}$ a

因此，我们运行完整优化两次：第一次（算法 1，第 30 行）有一个碰撞点 $\mathbf{x}$ c $=(\mathbf{b}$ a

$+\mathbf{b}$ b $)/2$, i.e., constrained to lie at the midpoint of the collision positions of both objects.

$+\mathbf{b}$ b $)/2$ ，即约束位于两个物体碰撞位置的中点。

Removing these degrees of freedom at first increases robustness, especially for cases with lower quality data.

首先删除这些自由度可以提高稳健性，特别是对于数据质量较低的情况。

For the second run, we release the collision point to compute the final parameters of the collision (Alg. 1, line 31).

对于第二次运行，我们释放碰撞点来计算碰撞的最终参数（算法 1，第 31 行）。

While mass ratio, c, etc., have been computed during the solve, we evaluate the final positions and orientations from our continuous, analytical parametrization.

虽然在求解过程中计算了质量比、c等，但我们通过连续分析参数化来评估最终位置和姿态。

In the following, we explain how to obtain measurements of screen space object centroids and 3D object orientations from the video.

下面，我们将解释如何从视频中获取屏幕空间对象质心和 3D 对象姿态的测量值。

4.2 Initializing Positions

4.2 位置初始化

We simultaneously estimate screen space object centroid measurements, assign them to trajectory segments, and initialize the collision in 3D.

我们同时估计屏幕空间对象质心测量值，将它们分配给轨迹段，并在 3D 中初始化碰撞。

We assume that both objects are roughly equidistant to the camera at all times and that their 3D bounding box estimates from the initial scans are available.

我们假设两个物体在任何时候都与相机大致等距，并且可以使用初始扫描得出的 3D 边界框估计值。

Essentially, we use the known gravitational acceleration and video frame rate to position the collision in 3D.

本质上，我们使用已知的重力加速度和视频帧速率来以 3D 方式定位碰撞。

Outline.

大纲。

First, we calculate image region candidates for the reprojection of the objects in each frame.

首先，我们计算每帧中对象重投影的图像区域候选。

Then, we perform a RANSAC labeling involving several, modified parabola fits to compute temporally consistent assignments of the image region centroids to the four trajectory segments (see line 10 in Alg. 1) and to identify false-positive image regions.

随后用 RANSAC 分配标签：执行多次修正的抛物线拟合，将图像区域质心按时间一致性分配给 4 个轨迹段（算法 1 第 10 行），并识别误检区域。

Finally, we iterate this process in a loop to determine the approximate time of collision.

最后，我们在循环中迭代此过程以确定碰撞的大致时间。

Image region calculation.

图像区域计算。

We estimate candidate center of mass positions in image space.

我们估计图像空间中的候选质心位置。

We use a standard background subtraction pipeline consisting of Gaussian mixture modeling [Zivkovic 2004] and connected contour detection [Teh and Chin 1989].

采用标准背景减除流程，包括高斯混合建模 [Zivkovic 2004] 和连通轮廓检测 [Teh and Chin 1989]。

Figure 4

图4

Algorithm 1: SMASH

算法1：SMASH

1 INITIALIZEPOSITIONS () /* Sec. 4.2 */

1 初始化位置 () /* 第 4.2 */

2 INITIALIZEORIENTATIONS () /* Sec. 4.3 */

2 初始化姿态 () /* 第 4.3 */

3 RECONSTRUCTCOLLISION () /* Sec. 4.1 */

3 重建碰撞 () /* 第 4.1 */

4 Function INITIALIZEPOSITIONS /* Sec. 4.2 */

4 函数 初始化位置 /* 第 4.2 */

5 Input :video, $\mathbf{s}^{\diamond}$**,** $\mathbf{P}$**,** f**fps,** g

5 输入 :video, $\mathbf{s}^{\diamond}$, $\mathbf{P}$, f fps, g

6 Assumption: $\beta_{y|}^{*}{}_{y0}^{*} = 0$, ba

6 假设: $\beta_{y|}^{*}{}_{y0}^{*} = 0$, b a

$\widetilde{t^c}$
7 d| c $\leftarrow N/2$

$\widetilde{t^c}$
7 d| c $\leftarrow N/2$

8 do

8 do

9 tc

9 t c

10 for e=1 to 25 do /* RANSAC loop */

10 for e=1 to 25 do /* RANSAC 循环 */

$\widehat{k=1}$ ← ASSIGNRND(SAMPLE({p 2d

$\widehat{k=1}$ ← ASSIGNRND(SAMPLE({p 2d

12 θ ← SOLVE({ pxy , siz , g } , {} , {p 2d ,*

12 θ ← 求解({ pxy , siz , g } , {} , {p 2d ,*

$\overset{N}{i=1}$ ← ASSIGNGREEDY({p 2d

$\overset{N}{i=1}$ ← ASSIGNGREEDY({p 2d

14 θ ← SOLVE({ pxy , siz , g } , {} , {p 2d ,*

14 θ ← 求解({ pxy , siz , g } , {} , {p 2d ,*

15 $\hat{\theta}$ ← COMPARE($\hat{\theta}$, SCORE(θ , {p 2d ,*

15 $\hat{\theta}$ ← COMPARE($\hat{\theta}$, SCORE(θ , {p 2d ,*

$\widetilde{t^c}$ $\widetilde{t^c}$
16 while ut tio c $\leq \lfloor t$**c** $_{init}^{c} \rfloor$ **or** $\{d_i$ **c** $\geq \lceil t$**c**

16 while $\tilde{t}^{\mathbf{ut}}_{\mathbf{tio}}\mathrm{c}^{\cdot} \leq \lfloor t^{\cdot}\mathrm{c}^{\mathrm{c}}_{init} \rfloor$ or $\tilde{t}^{\{d_{\cdot}}_{3\mathrm{O}}\mathrm{c}^{\cdot} \geq \lceil t^{\cdot}\mathrm{c}$

17 Output : $\{\mathbf{p2d}_{\cdot,*}{}^{\cdot}{}_{i}\}_{i=}^{N}$

17 输出 : $\{\mathbf{p}$ 2d $_{\cdot,*}{}^{\cdot}{}_{i}\}_{i=}^{N}$

18 Function INITIALIZEORIENTATIONS /* Sec. 4.3 */

18 函数 初始化姿态 /* 第 4.3 */

19 Input :video, $\{\mathbf{p}^{\cdot}\mathbf{2d}_{\cdot,*}$

19 输入 :video, $\{\mathbf{p}^{\cdot}$ 2d $_{\cdot,*}$

20 Output : $\{\mathbf{q}^{\cdot}\mathbf{obs}_{\cdot,*}{}^{\cdot}{}_{j}\}_{j=}^{M}$

20 输出 : $\{\mathbf{q}^{\cdot}$ obs $_{\cdot,*}{}^{\cdot}{}_{j}\}_{j=}^{M}$

$_{j=1}^{M} \leftarrow$ OPENDR(Input)

$_{j=1}^{M} \leftarrow$ OPENDR(输入)

21 Function RECONSTRUCTCOLLISION /* Sec. 4.1 */

21 函数 重建碰撞 /* 第 4.1 */

22 Input : $\{\mathbf{p2d}_{\cdot,*}$

22 输入 : $\{\mathbf{p}$ 2d $_{\cdot,*}$

$_{\mathbf{1}}{}^{\cdot}, {}^{\tilde{b}^{*}_{2}}_{b\}}{}^{\cdot}{}_{\mathrm{ost},\diamond}{}^{\cdot}{}_{\mathbf{2}}{}^{\cdot}, {}^{\tilde{\mathbf{b}}^{\diamond}_{3}}_{\mathbf{do}\}} \leftarrow {}^{\tilde{\beta}^{*}_{y}}_{\check{\mathbf{3}}}{}^{\cdot}{}_{j}{}^{\cdot} \mid {}^{\tilde{\beta}_{x}}_{\ddot{y}0}{}^{\cdot}, {}^{\tilde{\beta}_{y1}}_{\mathrm{obs},}{}^{\cdot}{}_{j}{}^{\cdot}, {}^{\cdot}\mathrm{clo}$ 23 for $^{\cdot}\diamond \in \{$ a $^{\cdot},$ b $^{\cdot}\}$ do

$_{\mathbf{1}}{}^{\cdot}, {}^{\tilde{b}^{*}_{2}}_{b\}}{}^{\cdot}{}_{\mathrm{ost},\diamond}{}^{\cdot}{}_{\mathbf{2}}{}^{\cdot}, {}^{\tilde{\mathbf{b}}^{\diamond}_{3}}_{\mathbf{do}\}} \leftarrow {}^{\tilde{\beta}^{*}_{y}}_{\check{\mathbf{3}}}{}^{\cdot}{}_{j}{}^{\cdot} \mid {}^{\tilde{\beta}_{x}}_{\ddot{y}0}{}^{\cdot}, {}^{\tilde{\beta}_{y1}}_{\mathrm{obs},}{}^{\cdot}{}_{j}{}^{\cdot}, {}^{\cdot}\mathrm{clo}$ 23 for $^{\cdot}\diamond \in \{$ a $^{\cdot},$ b $^{\cdot}\}$ do

24 $\{j$ pre $^{\cdot,\diamond}, j$ post $^{\cdot,\diamond}\} \leftarrow j \mid t^{\cdot}$ obs $_{\cdot,\diamond}$

24 $\{j$ pre $^{\cdot,\diamond}, j$ post $^{\cdot,\diamond}\} \leftarrow j \mid t^{\cdot}$ obs $_{\cdot,\diamond}$

j closest before/after $^{\cdot}t^{\cdot}$ c

j 之前/之后最近的 $^{\cdot}t^{\cdot}$ c

25 $\mathbf{q}^{\cdot}\mathbf{c}^{\cdot,\diamond} \leftarrow$SLERP($\mathbf{q}^{\cdot}\mathbf{obs}_{\cdot,\diamond}$

25 $\mathbf{q}^{\cdot}$ c $^{\cdot,\diamond} \leftarrow$ SLERP($\mathbf{q}^{\cdot}$ obs $_{\cdot,\diamond}$

$_{j}$ pre $^{\cdot,\diamond}_{\cdot,\diamond}, \mathbf{q}^{\cdot}$ obs $_{\cdot,\diamond}$

j pre $^{\diamond}$, $\mathbf{q}$ obs ,

j post $^{\diamond}$, t obs ,

j post $^{\diamond}$, t obs ,

26 for $j^* \in \{j$ pre $^{\diamond}, \check{j}$ post $^{\diamond}\}$ do

26 for $j^* \in \{j$ pre $^{\diamond}, \check{j}$ post $^{\diamond}\}$ do

$2(\mathbf{q}$ obs $,* \; j^* - \mathbf{q}$ c $^{\diamond})/(t$ obs

$2(\mathbf{q}$ obs $,* \; j^* - \mathbf{q}$ c $^{\diamond})/(t$ obs

28 $\mathbf{k}^* \leftarrow \mathbf{R}_{\mathbf{q}}\mathbf{c}_{,\diamond}\mathbf{I}_0\grave{\mathbf{R}}^{-1}_{\mathbf{a}^{\mathbf{c}},}$

28 $\mathbf{k}^* \leftarrow \mathbf{R}_{\mathbf{q}}$ c $_{,\diamond}\mathbf{I}_0\grave{\mathbf{R}}^{-1}_{\mathbf{a}^{\mathbf{c}},}$

29 $\mathbf{x}^{\mathbf{c}}$ $^{\mathbf{c}}_{init} \leftarrow (\widetilde{\mathbf{b}^{\mathbf{a}}_3}$ $\leftarrow$ SOLVE($\leftarrow$ SOLVE($\mathbf{b}^{\mathbf{a}}$

29 $\mathbf{x}^{\mathbf{c}}$ $^{\mathbf{c}}_{init} \leftarrow (\widetilde{\mathbf{b}^{\mathbf{a}}_3}$ $\leftarrow$ SOLVE($\leftarrow$ SOLVE($\mathbf{b}^{\mathbf{a}}$

30 $\theta \leftarrow$ SOLVE({ mom , imp , g , ke , CoR , pxy , pz , ori }, $\{\mathbf{x}^{\mathbf{c}}$ })

30 $\theta \leftarrow$ 求解({ mom , imp , g , ke , CoR , pxy , pz , ori }, $\{\mathbf{x}^{\mathbf{c}}$ })

31 $\theta \leftarrow$ SOLVE({ mom , imp , g , ke , CoR , pxy , pz , ori }, {})

31 $\theta \leftarrow$ 求解({ mom , imp , g , ke , CoR , pxy , pz , ori }, {})

32 Output : $b_1^{*}, \grave{b}_2^{*}, \mathbf{b}_3^{\diamond}, \beta_{y'}^{*}, \beta_{y0}^{*}, \beta_x, \beta_{y1}, \mathbf{q}\mathbf{c}^{\diamond}, \hat{\mathbf{k}}^{*}$

32 输出 : $b_1^{*}, \grave{b}_2^{*}, \mathbf{b}_3^{\diamond}, \beta_{y'}^{*}, \beta_{y0}^{*}, \beta_x, \beta_{y1}, \mathbf{q}$ c $^{\diamond}, \hat{\mathbf{k}}^{*}$

33 Function SOLVE (Terms, FixedVars, Data = All)

33 函数 SOLVE (能量项, 固定变量, 数据 = 全部)

34 return θ = solve Eq. (15) | Terms, FixedVars, Data

34 返回 θ = 求解式 (15) | 能量项 , 固定变量 , 数据

shows the output of the background subtraction step for three frames of a collision.

显示了碰撞三帧的背景扣除步骤的输出。

RANSAC labeling.

RANSAC 标签。

We choose the middle of the video sequence as initial guess for the collision time t_c, more specifically, the midpoint in time between the two consecutive images.

我们选择视频序列的中间作为碰撞时间t_c的初始猜测，更具体地说，是两个连续图像之间的时间中点。

We then assign a randomly chosen centroid position from each video frame before and after t_c to the pre- and post-collision parabolas respectively (line 11 in Alg. 1).

然后，我们将t_c之前和之后的每个视频帧中随机选择的质心位置分别分配给碰撞前和碰撞后抛物线（算法1中的第11行）。

We omit f fps $/10$ frames (24 at 240fps) around collision time, as we found these to be particularly unreliable due to the merged silhouettes.

我们省略了碰撞时间附近的$f$fps$/10$帧（24，240fps），因为我们发现由于合并的轮廓，这些帧特别不可靠。

We perform a modified non-linear least squares fit to calculate the parabola parameters from Section 3.3 using the assigned screen space centroids, only considering the 2D position fidelity term (Equation (22)) and a new size penalty term:

我们使用指定的屏幕空间质心执行修改的非线性最小二乘拟合来计算第 3.3 节中的抛物线参数，仅考虑 2D 位置保真度项（方程（22））和新的尺寸惩罚项：

$$f^{\text{siz}} = \frac{1}{N}\sum_{i=1}^{N}\left(\|\mathbf{P}(\mathbf{p}^*(t_i)+\frac{\mathbf{d}}{2}) - \mathbf{P}(\mathbf{p}^*(t_i)-\frac{\mathbf{d}}{2})\| - s_i^{2\text{d}}\right)^2, \quad (26)$$

where $\mathbf{p}^*(t)$ is the position on a parabola at time t, $\mathbf{d}$ is the 3D

其中$\mathbf{p}^*(t)$是t时刻抛物线上的位置，$\mathbf{d}$是3D

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diagonal of the scanned object, s 2d denotes the size of a circle fit to the image region corresponding to the selected centroid for this frame.

扫描对象的对角线，s 2d 表示适合与该帧所选质心相对应的图像区域的圆的大小。

This term pushes a parabola position towards a depth value so that the projected object diagonals match the image region sizes.

该项将抛物线位置推向深度值，以便投影的对象对角线与图像区域大小匹配。

For this optimization, we share the z-coordinate of $\mathbf{b}_3$ between the two objects to ensure the objects meet in space.

对于此优化，我们在两个对象之间共享 $\mathbf{b}_3$ 的 z 坐标，以确保对象在空间中相遇。

We also fix the parabola rotations around y to 0 (β^*_{y0}

我们还将绕 y 的抛物线旋转固定为 0 （β^*_{y0}

$= 0$) to stabilize the non-linear solve by constraining the space of solutions to be parallel to the camera plane.

$= 0$) 通过将解空间约束为与相机平面平行来稳定非线性求解。

The subsequent main optimization step will refine these estimates.

随后的主要优化步骤将完善这些估计。

The parabola fit allows us to greedily assign centroids to trajectory segments or label them as outliers (line 13 in Alg. 1).

抛物线拟合允许我们贪婪地将质心分配给轨迹段或将它们标记为异常值（算法 1 中的第 13 行）。

To evaluate the quality of the obtained fit, we compute a score based on the residual of the parabola fit optimization, in conjunction with a SIFT feature distance f lm to measure the similarity of the selected image regions.

为了评估所获得拟合的质量，我们根据抛物线拟合优化的残差计算分数，并结合 SIFT 特征距离 f lm 来测量所选图像区域的相似性。

For the former, we use the screen-space, area-weighted centroid-to-parabola distances d_i para from Eq. (22), multiplied with f inlier, the ratio of centroids correctly explained (with an inlier distance equal to the median size of the identified regions).

前者使用式（22）中按区域面积加权的图像平面质心到抛物线距离 d_i para，再乘以被正确解释的质心比例 f inlier；内点距离阈值取已识别区域尺寸的中位数。

The image content score is calculated with a SIFT descriptor distance of the object regions between consecutive frames.

图像内容得分是通过连续帧之间的对象区域的 SIFT 描述符距离来计算的。

The final score is weighted sum of these terms: d_i para f inlier $+ w$ lm f lm, see line 15 in Alg. 1.

最终分数是这些项的加权和： $d_i$para $f$inlier $+ w$lm $f$lm，参见算法中的第15行。 1.

The fit with the lowest score is used to store the final set of screen-space centroids $\{\mathbf{p}$ 2d $,*$

得分最低的拟合用于存储最终的屏幕空间质心集 $\{\mathbf{p}$ 2d $,*$

$i=1$, which are passed on to the main optimization stage.

$i=1$ ，将被传递到主优化阶段。

In addition to the 2D positions, we store the depth $\{d_i^*$

除了 2D 位置之外，我们还存储深度 $\{d_i^*$

$i=1$ along each of the view rays for our depth terms (Eq. (23)), and their assignments to the trajectory segments.

$i=1$沿着我们的深度项（方程（23））的每条视图光线，以及它们对轨迹段的分配。

Figure 5 visually illustrates the steps of a single RANSAC iteration, typically converging in about 25 iterations.

图 5 直观地说明了单个 RANSAC 迭代的步骤，通常在大约 25 次迭代中收敛。

Collision time refinement.

碰撞时间细化。

Note that the optimization above computes an improved value for t c based on the intersection of the parabolas in Eq. (22).

上述优化根据式（22）中抛物线的交点，得到更准确的碰撞时刻 t c。

Initially, t c is set to the midpoint of a pair of video frames.

最初， tc被设置为一对视频帧的中点。

If t c shifts before the first image of the pair, or to a point in time after the second one, we restart the calculation after selecting the previous, or next pair of frames, respectively.

如果 tc移动到该对的第一幅图像之前，或者移动到第二幅图像之后的某个时间点，我们分别在选择上一帧或下一对帧后重新开始计算。

Their midpoint is the new initial guess for t c, and we re-run the outlier detection, as described above.

它们的中点是 t_c 的新初始猜测，我们重新运行异常值检测，如上所述。

As this optimization runs very fast, we found that a sequential search for the correct collision time along the video works very well.

由于这种优化运行得非常快，我们发现沿着视频顺序搜索正确的碰撞时间效果非常好。

Alternatively, a bisection search could be easily implemented.

或者，可以轻松实现二分搜索。

4.3 Initializing Orientations

4.3 初始化姿态

The previous initialization step yields a first set of 3D positions in time.

前面的初始化步骤及时生成了第一组 3D 位置。

We can render the scanned objects over the video on these

可以沿这些轨迹，将扫描所得物体渲染并叠加到视频上，

(a) image region

(a) 图像区域

openDR #1 openDR #2 openDR #3

OpenDR #1 OpenDR #2 OpenDR #3

(b) image region

(b) 图像区域

openDR #1 openDR #2 openDR #3

OpenDR #1 OpenDR #2 OpenDR #3

Figure 6: Two examples of the image content (left) and the resulting candidate orientations (right).

图 6：图像内容（左）和生成的候选姿态（右）的两个示例。

Green check marks highlight the selected orientations.

绿色复选标记突出显示所选姿态。

top view top view

顶视图 顶视图

Figure 7: a) the result of our initialization: labeled centroids and approximate depth values (green dots) together with the initialized orientations (green cuboids); b) 3D parabolas resulting from our full optimization (red curves), and c) motion over time (colored cuboids).

图 7：a) 初始化的结果：标记的质心和近似深度值（绿点）以及初始化的姿态（绿色长方体）； b) 我们全面优化产生的 3D 抛物线（红色曲线），以及 c) 随着时间的推移的运动（彩色长方体）。

trajectories to initialize the orientations, assuming that 3D scans of the filmed objects are available for this step.

以初始化姿态；此步骤假定已获得所拍摄物体的三维扫描。

Using a differential renderer [Loper and Black 2014] and the object scans, we perform a first alignment of each object starting from 512 viewpoints uniformly sampled on a sphere.

利用可微渲染器 [Loper and Black 2014] 和物体扫描模型，从球面上均匀采样的 512 个视角出发，为各物体进行初步对齐。

However, due to illumination variations we found the video information to be typically too ambiguous to automatically retrieve the correct orientations for all cases.

然而，由于照明变化，我们发现视频信息通常过于模糊，无法自动检索所有情况下的正确姿态。

Hence, we let a user check, and if necessary correct these alignments for two points in time for each parabola segment.

因此，我们让用户检查，并在必要时纠正每个抛物线段的两个时间点的对齐情况。

This step takes less than a minute per collision sequence.

每个碰撞序列的此步骤花费不到一分钟。

An example of the generated candidate orientations can be seen in Figure 6. The labelled screen-space centroid positions $\{\mathbf{p}$ 2d $\}_{,*}$

生成的候选姿态的示例如图 6 所示。 标记的屏幕空间质心位置 $\{\mathbf{p}$ 2d $\}_{,*}$

$_{i=1}$ with depths $\{d_i^*$

$_{i=1}$有深度$\{d_i^*$

$_{i=1}$, and the $M = 8$ input orientations $\{\mathbf{q}$ obs $\}_{,*}$

$i=1$、$M = 8$输入姿态 $\{\mathbf{q}$ obs $_{i,*}$

$j=1$ are the input data for our full parameter estimation step (Section 4.1).

$j=1$是我们完整参数估计步骤（第4.1节）的输入数据。

4.4 Implementation

4.4 实施

We use CERES [Agarwal et al. 2016] for solving our non-linear least squares formulation, with sparse-Cholesky as linear solver type, and energy derivatives from automatic differentiation.

我们使用 CERES [Agarwal 等人。 2016] 用于求解非线性最小二乘公式，使用稀疏 Cholesky 作为线性求解器类型，以及自动微分的能量导数。

Figure 7 shows an example result of the initialization, and the objects' recovered 3D positions and orientations.

图 7 显示了初始化的示例结果以及对象恢复的 3D 位置和姿态。

Due to the non-linearity of the angular motion in Eq. (8) we typically use four integration steps per frame.

由于方程中角运动的非线性。 (8) 我们通常每帧使用四个积分步骤。

Note that the sequence of integration steps is taken into account during the automatic differentiation performed in the solver.

请注意，在求解器中执行自动微分期间会考虑积分步骤的顺序。

While higher-order integrators or even closed-form Poinsot solutions would be available to express a rigid body orientation over time, we found that explicit integration yields stable solutions in practice, and leave alternatives for future work.

虽然高阶积分器甚至闭式 Poinsot 解决方案可用于表达随时间变化的刚体姿态，但我们发现显式积分在实践中会产生稳定的解决方案，并为未来的工作留下替代方案。

We use f fps to convert time t to be measured in frames.

我们使用 $f$fps来转换 t以帧为单位测量的时间。

In our formulation above, we took care to minimize the number of free variables and energy terms.

在上面的公式中，我们注意尽量减少自由变量和能量项的数量。

It turned out to be crucial for a good solution to use a formulation with shared variables, instead of enforcing equalities with highly weighted penalty terms.

事实证明，对于一个好的解决方案来说，使用具有共享变量的公式至关重要，而不是通过高权重惩罚项来强制执行等式。

For example, instead of adding a term to minimize b pre $,$ a

例如，不要添加一项来最小化 b之前的$,$a

3 , we formulate the parabolas with a shared variable, which inherently enforces the equality of the offsets.

3，我们用共享变量制定抛物线，这本质上强制了偏移量的相等。

Likewise, equivalent angles and orientations are implemented with shared variables instead of

同样，等效角度和姿态是通过共享变量而不是

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Scene (#frames) BgFg IniPos IniOri Solve 1 Solve 2 box-1 (217) 5.68s 2m 6s 18m 46s 0.55s 0.08s box-2 (191) 4.51s 1m 51s 17m 47s 0.23s 0.03s box-3 (236) 8.12s 2m 0s 19m 2s 0.36s 0.05s helmet-duck (316) 8.63s 1m 40s 24m 41s 0.98s 0.12s duck-eleph.

场景 (#frames) BgFg IniPos IniOri 求解 1 求解 2 box-1 (217) 5.68s 2m 6s 18m 46s 0.55s 0.08s box-2 (191) 4.51s 1m 51s 17m 47s 0.23s 0.03s box-3 (236) 8.12s 2m 0s 19m 2s 0.36s 0.05s 头盔鸭 (316) 8.63s 1m 40s 24m 41s 0.98s 0.12s 鸭大象。

(204) 5.53s 1m 57s 25m 55s 3.08s 0.18s duck-rugby (316) 11.86s 1m 43s 25m 11s 0.65s 0.36s

(204) 5.53s 1m 57s 25m 55s 3.08s 0.18s 鸭式橄榄球 (316) 11.86s 1m 43s 25m 11s 0.65s 0.36s

Table 2: Timings on a system with Intel i7-4700MQ and Nvidia Geforce GTX 770M.

表 2：在 Intel i7-4700MQ、Nvidia GeForce GTX 770M 系统上各阶段的运行耗时。

The second and third columns describe Section 4.2, the fourth column Section 4.3.

第二列和第三列描述第 4.2 节，第四列描述第 4.3 节。

The rightmost two columns reference lines 30 and 31 of Algorithm 1.

最右边的两列参考算法 1 的第 30 行和第 31 行。

additional penalty terms.

额外惩罚项来实现。

See Table 2 for runtimes.

有关运行时间，请参阅表 2。

We have made code and data available online1.

我们已在线提供代码和数据1。

5 Evaluation

5 评价

We tested the robustness of our method and quantitatively evaluated the accuracy of the estimates.

我们测试了我们方法的稳健性，并定量评估了估计的准确性。

First, we report performance on synthetic data to test the system under different perturbations (Section 5.1) and then evaluate the system on a range of real-world sequences of increasing complexity (Section 5.2).

首先，我们报告合成数据的性能，以在不同扰动下测试系统（第 5.1 节），然后在一系列复杂性不断增加的现实世界序列上评估系统（第 5.2 节）。

Accuracy measures.

精度指标。

We evaluated the results as: (i) in a synthetic setup, we compare c and $m^{\prime}$ b,a estimates against ground truth values used to generate the test sequences; (ii) in real-world setups, we either compare with estimates obtained via other means (e.g., using potential energy), or by evaluating consistency of the estimates across multiple collision sequences involving same object pairs.

我们将结果评估为：（i）在综合设置中，我们将c和mb,a估计值与用于生成测试序列的真值进行比较；（ii）在现实世界的设置中，我们要么与通过其他方式（例如，使用势能）获得的估计进行比较，要么通过评估涉及相同对象对的多个碰撞序列的估计的一致性。

We use $\tilde{x}$ to denote our estimated value for quantity x, thus an output of the optimization.

我们使用$\tilde{x}$来表示我们对数量x的估计值，即优化的输出。

E.g., $\tilde{c}$ for our estimate of a coefficient of restitution c.

例如，$\tilde{c}$用于我们对恢复系数的估计c。

Where applicable, ground truth values measured by other means will be denoted with an m subscript, e.g., c_m .

在适用的情况下，通过其他方式测量的真值将用m下标表示，例如c_m。

Weights.

权重。

In all the following tests we used a fixed set of weights: w mom $= w$ imp $= 10$, w g $= w$ ke $= w$ CoR $= w$ ori $= w$ pxy $= 1$, w pz $= 0.1$, $\epsilon = 1$, and for the initialization phase w siz $= 0.1$, w Im $= 1/500$.

在以下所有测试中，我们使用了一组固定的权重：w mom $= w$ imp $= 10$, w g $= w$ ke $= w$ CoR $= w$ ori $= w$ xy $= 1$、$w$pz $= 0.1$、$\epsilon = 1$，以及初始化阶段$w$siz $= 0.1$, wIm $= 1/500$。

5.1 Validation on Synthetic Data

5.1 合成数据验证

We tested our system using synthetic data obtained from forward simulations using the BULLET physics engine.

我们使用 BULLET 物理引擎的正向模拟获得的合成数据测试了我们的系统。

We created a scene with two colliding cuboids under gravity and rendered the simulation as input video for our algorithm.

我们创建了一个包含两个在重力作用下碰撞的长方体的场景，并将模拟渲染为我们算法的输入视频。

(i) Robustness to initialization errors.

(i) 对初始化错误的鲁棒性。

We separately tested the effect of errors in 2D centroid and orientation estimation in the initialization stage.

我们在初始化阶段分别测试了二维质心和姿态估计误差的影响。

We rendered the synthetic sequence at 60fps, initialized position for each time frame (Section 4.2), and added increasing amount of uniform noise to the retrieved 2D object centroids before fitting the parabolas.

我们以 60fps 渲染合成序列，初始化每个时间帧的位置（第 4.2 节），并在拟合抛物线之前向检索到

的 2D 对象质心添加越来越多的均匀噪声。

The algorithm provided robust estimates for both c and m b,a up to perturbation margins of 30% (Figure 8(a)) with the noise margin being compared to the average screen-space size of the larger object.

该算法为 c和 mb,a提供了稳健的估计，直至30%的扰动裕度（图8（a）），并将噪声裕度与较大物体的平均屏幕空间尺寸进行比较。

We ran 63 experiments with different noise margins ($0-30\%$) and found that our parabola fitting to be robust even under moderate errors on centroid estimates.

我们用不同的噪声容限（$0-30\%$）进行了 63 次实验，发现即使在质心估计存在中等误差的情况下，我们的抛物线拟合也是稳健的。

For perturbations involving orientation annotations (Section 4.3), we added noise to each coordinate, and re-normalized the resultant quaternions.

对于涉及姿态标注的扰动（第 4.3 节），我们向每个坐标添加噪声，并对所得四元数重新归一化。

We observed robustness in both $\tilde{c}$ and $\tilde{m}$ b,a up to perturbation of around 40° around actual orientation estimates (Figure 8(b)).

我们观察到 $\tilde{c}$和 $\tilde{m}$b,a 的稳健性，直到围绕实际姿态估计的40°附近的扰动（图8（b））。

1http://geometry.cs.ucl.ac.uk/projects/2016/smash/

脚注 1：http://geometry.cs.ucl.ac.uk/projects/2016/smash/

(a) Robustness to perturbations in positional estimates

(a) 位置估计中对扰动的鲁棒性

relative mass ratio

相对质量比

CoR estimate CoR estimate

CoR 估计 CoR 估计

uniform noise (% of blob diameter) uniform noise (% of blob diameter)

均匀噪声（斑点直径的%） 均匀噪声（斑点直径的%）

(b) Robustness to perturbations in orientation estimates

(b) 姿态估计中对扰动的鲁棒性

relative mass ratio

相对质量比

uniform noise added to orientation quaternions

添加到姿态四元数的均匀噪声

uniform noise added to orientation quaternions

添加到姿态四元数的均匀噪声

Figure 8: Synthetic evaluation of robustness with increasing noise in the position and orientation estimates.

图 8：随着位置和姿态估计中噪声的增加，鲁棒性的综合评估。

Green lines denote ground truth, $c = 0.49$ and $m_b/m_a = 1.33$.

绿线表示真值，$c = 0.49$和$m_b/m_a = 1.33$。

CoR estimate

CoR估计

all physics priors

所有物理先验

impulse term only conservation terms only

仅冲量项 仅守恒项

CoR in [0,1] only KE bound only

仅限制恢复系数在 [0,1]；仅限制动能上界

no physics (only shared variables)

没有物理（只有共享变量）

estimated relative mass

估计相对质量

Figure 9: Evaluation of the role of different terms.

图 9：各能量项作用的评估。

Impulse-based physics coupling is clearly the essential ingredient, while the other terms act as secondary regularizer.

基于冲量的物理耦合显然是基本成分，而其他项则充当辅助正则化器。

Larger circles show experiments with larger noise (positions and orientations).

较大的圆圈表示具有较大噪声（位置和姿态）的实验。

The input was a noise-free run of our automatic parabola initialization and synthetic orientations at least five timesteps away from collision time.

输入是我们的自动抛物线初始化和合成姿态的无噪声运行，距碰撞时间至少五个时间步长。

Dashed lines denote ground truth, $c = 0.49$ and m b,a $= 1.33$.

虚线表示真值，$c = 0.49$和mb,a$= 1.33$。

(ii) Effect of physics prior.

(ii) 物理先验的影响。

Next, in an important test, we assessed the contribution of the different physics-based priors in regularizing the optimization.

接下来，在一项重要的测试中，我们评估了不同的基于物理的先验在正则化优化方面的贡献。

We experimented by activating a selection of the optimization terms and evaluating estimation accuracy (see Figure 9).

我们通过激活优化项的选择并评估估计准确性进行了实验（见图 9）。

The impulse term has the single most important effect: while just based on the input video, the estimates for c and m b,a deviate substantially, the impulse-based physics formulation provides valuable regularization resulting in reliable estimates even in the noisy setting.

冲量项具有最重要的影响：虽然仅基于输入视频，c和mb,a 的估计值存在很大偏差，但基于冲量的物理公式提供了有价值的正则化，即使在含噪声的环境中也能产生可靠的估计。

The secondary constraints in the form of bounds on c and kinetic energy further improve the estimation accuracy.

c和动能上的边界形式的二次约束进一步提高了估计精度。

(iii) Effect of annotation frequency.

(iii)标注频率的影响。

We measured the effect of fewer number of orientation annotations across time on estimated m b,a and c .

我们测量了随着时间的推移，较少数量的姿态标注对估计的mb，a和c的影响。

Figure 11 shows the estimates for increasing annotation intervals.

图 11 显示了增加标注间隔的估计。

Note that even for the extreme case with an interval of around 20, i.e., with only 4 annotations for a 90 frame sequence our

请注意，即使对于间隔约为 20 的极端情况，即 90 帧序列只有 4 个标注，我们的

solver computes reliable estimates.

求解器计算可靠的估计。

In all the other experiments, we used 2 orientation annotations for each of the 4 parabolas for a pair of colliding objects.

在所有其他实验中，我们对一对碰撞物体的 4 条抛物线中的每一条都使用了 2 个姿态标注。

(iv) Effect of friction.

(iv) 摩擦力的影响。

Recall that in our formulation, we ignored the effect of friction.

回想一下，在我们的公式中，我们忽略了摩擦的影响。

Hence, we evaluated the effect of ignoring friction on our estimates.

因此，我们评估了忽略摩擦对我们的估计的影响。

In the experiments, we changed the (static) friction coefficient and coefficient of restitution of the simulator and re-ran the simulation.

在实验中，我们改变了模拟器的（静）摩擦系数和恢复系数并重新运行模拟。

We then estimated c and m b,a using our algorithm on the simulation output.

然后，我们使用我们的算法对模拟输出估计c和mb,a。

As shown in Figure 10, we observed that while the relative mass estimates continued to be good, the coefficient of restitution was underestimated.

如图 10 所示，我们观察到，虽然相对质量估计值仍然不错，但恢复系数被低估了。

This is explained by our model ignoring energy loss due to friction, especially when the friction coefficient is (synthetically) increased.

这是因为模型忽略了摩擦造成的能量损失，尤其是在仿真中人为增大摩擦系数时。

5.2 Testing on Real Sequences

5.2 真实序列测试

We captured the real-world sequences with 1280×720 video with 120 fps or 240 fps using a regular iPhone6 with known camera intrinsics.

使用内参已知的普通 iPhone 6，以 1280×720 分辨率、120 fps 或 240 fps 拍摄真实序列。

The colliding objects were scanned in static positions using Kinect Fusion.

使用 Kinect Fusion 在静态位置扫描碰撞物体。

Potential energy comparison.

势能比较。

One way to measure the coefficient of restitution is based on the ratio of energy loss during a collision.

测量恢复系数的一种方法是基于碰撞期间能量损失的比率。

However, this requires a very controlled setting, where we can measure the change of energy using only the objects' potential energy, i.e., when they have zero kinetic energy.

然而，这需要一个非常受控的设置，我们可以仅使用物体的势能（即当它们的动能为零时）来测量能量的变化。

We use different balls dropped from a controlled height h, measuring the apex of their trajectories after they bounce off the floor.

我们使用从受控高度落下的不同球 h，测量它们从地面弹起后的轨迹顶点。

When no energy is transferred into an angular motion, we can compute $c =$

当没有能量转化为角运动时，我们可以计算$c =$

h_i after $/h_i$ before.

h之后/h之前。

We experimented using four different balls (ping-pong, tennis, football, and rugby) and computed the resulting c values.

我们使用四种不同的球（乒乓球、网球、足球和橄榄球）进行实验，并计算了结果c值。

We also used our algorithm to independently estimate c .

我们还使用我们的算法独立估计c。

As this setup involves one static object (the floor), we remove all unknowns of the second object from our solve.

由于此设置涉及一个静态对象（地面），因此我们从求解中删除第二个对象的所有未知数。

We took 3 sets of measurements for each object.

我们对每个物体进行了 3 组测量。

Our estimates $\tilde{c}$ and the values measured with potential energy loss c_m are listed in the table in Figure 12. Note that for the examples without any spin, the algorithm ran in an automatic mode as we do not require any orientation annotation.

我们的估计 $\tilde{c}$和势能损失测量的值 c_m列在图12的表中。请注意，对于没有任何旋转的示例，算法以自动模式运行，因为我们不需要任何姿态标注。

Our method continues to consistently estimate c even when some energy is stored/converted to kinetic energy throughout the recording, i.e., the ball is spinning, or it starts off spinning (Figure 13).

我们的方法继续一致地估计 c，即使在整个记录过程中一些能量被存储/转换为动能，即球正在旋转，或者它开始旋转（图13）。

The simple estimation with potential energy is not valid for this case, while our formulation is still able to compute an estimate very close to the non-spinning case.

势能的简单估计对于这种情况无效，而我们的公式仍然能够计算出非常接近非旋转情况的估计。

Orientation annotations were provided for the scenes rug-1 and rug-2.

为场景 rug-1 和 rug-2 提供了姿态标注。

Please refer to the supplementary video for comparison.

请参阅补充视频进行比较。

relative change in CoR compared to no friction (%)

与无摩擦相比 CoR 的相对变化 (%)

CoR estimate

CoR估计

BLENDER’s Coulomb friction coefficient

BLENDER 的库仑摩擦系数

BLENDER’s Coulomb friction coefficient

BLENDER 的库仑摩擦系数

Figure 10: Evaluation of the role of friction.

图 10：摩擦作用的评估。

We changed the friction coefficient and coefficient of restitution in the input simulation, and ran our algorithm on the simulator’s output log (no noise).

我们改变了输入模拟中的摩擦系数和恢复系数，并在模拟器的输出日志上运行我们的算法（无噪音）。

The c value estimates decrease a bit, which is expected, since some energy is lost to friction.

c 值估计值略有下降，这是预期的，因为一些能量因摩擦而损失。

Solid lines on left denote ground truth.

左边的实线表示真值。

5.3 Real-world Cases

5.3 现实案例

Collisions between box-like object pairs.

盒状物体对之间的碰撞。

We next focus on collisions between pairs of freely moving objects.

接下来我们关注自由移动物体对之间的碰撞。

In contrast to the potential energy cases, there is no way to measure a ground truth value c_m and we only present the value calculated with our method.

与势能情况相反，没有办法测量真值 c_m，我们只呈现用我们的方法计算的值。

We can however measure ground truth for the objects' mass ratio.

然而，我们可以测量物体质量比的真值。

$\underline{m}$ box-1 0.28 0.91 1.0 box-2 0.39 1.21 1.0 box-3 0.20 1.0 1.0

$\underline{m}$ 盒-1 0.28 0.91 1.0 盒-2 0.39 1.21 1.0 盒-3 0.20 1.0 1.0

Table 3: Results of different collisions of cardboard boxes.

表 3：纸板箱不同碰撞的结果。

Table 3 lists our estimates across three captured videos of the boxes colliding in different situations.

表 3 列出了我们对三个捕获的盒子在不同情况下碰撞的视频的估计。

Note that in all these examples, the video sequences around (estimated) collision times are ignored by our method as those frames did not provide any reliable position/orientation estimates due to overlapping objects.

请注意，在所有这些示例中，我们的方法会忽略（估计的）碰撞时间周围的视频序列，因为这些帧由于重叠的对象而没有提供任何可靠的位置/姿态估计。

While the mass ratios are stable, and especially close to the ground truth for the last run, the $\tilde{c}$ vary more strongly.

虽然质量比稳定，并且特别接近上次运行的真值，但 $\tilde{c}$ 变化更大。

We attribute this partially to the fact that the cardboard boxes exhibit different stiffnesses depending on where they collide (based on the internal structure of the glued paper), in addition to the noise of the centroids estimated from the video.

我们将其部分归因于这样一个事实：除了从视频中估计的质心的噪声之外，纸板箱根据碰撞位置（基于胶合纸的内部结构）表现出不同的刚度。

Figure 14(a)-(c) shows some frames from the box sequences.

图 14(a)-(c) 显示了框序列中的一些帧。

Effect of direct keyframe-interpolation.

直接关键帧插值的效果。

We evaluate the effect of position and orientation interpolation while ignoring the underlying physics of collision.

我们评估位置和姿态插值的效果，同时忽略潜在的碰撞物理原理。

For this test, we used the initial position and orientation estimates obtained in our initialization stage as keyframes.

对于此测试，我们使用在初始化阶段获得的初始位置和姿态估计作为关键帧。

As shown in Figure 15-middle, such a direct interpolation without access to physics-based constraints resulted in ‘missing’ the actual collision, i.e., the duck and elephant ‘separated’ without actually colliding.

如图 15 中所示，这种没有基于物理约束的直接插值导致“错过”实际碰撞，即鸭子和大象“分离”而没有实际碰撞。

In comparison, Figure 15-bottom shows that our method accurately reconstructs the collision.

相比之下，图 15 底部显示我们的方法准确地重建了碰撞。

Collisions between non-box objects.

非盒子对象之间的碰撞。

We next evaluate how our algorithm performs on collision videos involving non-box objects.

接下来我们评估我们的算法在涉及非盒子对象的碰撞视频上的表现。

We present three sequences of varying complexity: helmet-duck, duck-elephant, and duck-rugby.

我们提出了三个不同复杂性的序列：头盔-鸭子、鸭子-大象和鸭子-橄榄球。

Table 4 lists estimation results.

表4列出了估算结果。

Figure 14(d)-(e) show the first two results.

图 14(d)-(e) 显示了前两个结果。

As the frame-level comparison shows the reconstructed motion, both in terms of position and orientation, is visually very accurate.

帧级比较表明，重建的运动在位置和姿态方面在视觉上都非常准确。

Although we validated the estimated mass ratios, we could not validate the accuracy of the estimated $\tilde{c}$ values.

尽管我们验证了估计的质量比，但我们无法验证估计的$\tilde{c}$值的准确性。

Qualitatively the $\tilde{c}$ estimates seem plausible: for example, the bike helmet gives a low estimate as it acts as a shock absorber; in contrast, the elephant and the rugby are bouncy and give high $\tilde{c}$ estimates, with the rugby giving the highest value.

从定性上看，$\tilde{c}$估计值似乎是合理的：例如，自行车头盔的估计值较低，因为它充当减震器；相比之下，大象和橄榄球有弹性，并给出较高的$\tilde{c}$估计值，其中橄榄球给出的值最高。

$\underline{m}$ helmet-duck 0.17 0.81 1.0 duck-elephant 0.46 2.33 2.0 duck-rugby (Fig. 1) 0.72 1.43 1.65

$\underline{m}$ 头盔-鸭子 0.17 0.81 1.0 鸭子-大象 0.46 2.33 2.0 鸭子-橄榄球（图1） 0.72 1.43 1.65

Table 4: Estimates of internal physics properties ($\tilde{c}$ and $\tilde{m}$ b,a) of complex objects.

表 4：复杂物体内部物理特性的估计（$\tilde{c}$和$\tilde{m}$b,a）。

The helmet clearly dissipates the most energy, whilst a sports ball is the most elastic by design.

显然，头盔消耗的能量最多，而运动球的设计弹性最大。

The duck-rugby sequence as shown in Figure 1 is the most challenging example as the objects remain hidden behind the curtain in the video for a large time interval.

图 1 所示的鸭子橄榄球序列是最具挑战性的示例，因为视频中的物体在很长一段时间内都隐藏在幕布后面。

Still the reconstruction

即便如此，重建

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uniform noise added to orientation quaternions uniform noise added to orientation quaternions

添加到姿态四元数的均匀噪声 添加到姿态四元数的均匀噪声

Figure 11: Synthetic evaluation of robustness towards noise in the orientations.

图 11：针对姿态噪声的鲁棒性综合评估。

Gap (2, 4, 6, 8, 10) denotes the distance in time to the closest orientation around collision time.

间隙 (2, 4, 6, 8, 10) 表示到碰撞时间附近的最近姿态的时间距离。

Dashed lines denote ground truth, $c = 0.49$ and m b,a $= 1.33$.

虚线表示真值，$c = 0.49$和mb,a $= 1.33$。

(mean and +/- std over 10 samples each, 2100 experiments overall)

每种设置取 10 个样本，报告均值和 ± 标准差；共开展 2100 次实验。

CoR estimate

CoR估计

results closely follow the recorded video.

结果仍与录制视频高度一致。

Note that this example is particularly challenging since the duck is comparatively small and featureless, making the initial orientation estimates noisy.

请注意，这个例子特别具有挑战性，因为鸭子相对较小且没有特征，使得初始姿态估计充满噪音。

5.4 Authoring

5.4 创作

We developed a BLENDER plugin (see Figure 16) to help author complex novel collision sequences using our parameter estimates from real collisions.

我们开发了一个 BLENDER 插件（见图 16），以帮助使用我们从实际碰撞中估计的参数来创作复杂的新颖碰撞序列。

The plugin allows the user to combine several observed collisions, time them to synchronize, or even stagger the interactions (i.e., collisions).

该插件允许用户组合多个观察到的碰撞，对它们进行同步计时，甚至交错交互（即碰撞）。

We used BULLET to obtain the object trajectories as accurately as possible using our estimated collision parameters.

我们使用 BULLET 使用估计的碰撞参数尽可能准确地获取对象轨迹。

Note that the user does not have to guess physical parameters for objects (i.e., c and m b,a values).

请注意，用户不必猜测对象的物理参数（即c和mb，a值）。

In Figure 16 we show the construction of a complex case of three pairs of rigid bodies colliding in mid air and two of them colliding again afterwards on their downward trajectories.

在图 16 中，我们展示了三对刚体在半空中碰撞的复杂情况的构造，其中两对刚体随后在向下的轨迹上再次碰撞。

Based on our reconstructed solutions, the new simulations behave naturally, and the objects collide as we would expect them to do from their real-world counterparts.

基于我们重建的解决方案，新的模拟行为自然，并且对象发生碰撞，正如我们期望它们与现实世界中的对应对象一样。

scene $\tilde{c}$ c m pong-0 0.55 pong-1 0.57 0.54 pong-2 0.55 tennis-0 0.71 tennis-1 0.73 0.76 tennis-2 0.71 football-0 0.66 football-1 0.67 0.77 football-2 0.67 rug-0 0.80 0.82 rug-1 (spin) 0.80 - rug-2 (spin) 0.78 -

场景 $\tilde{c}$ c m pong-0 0.55 pong-1 0.57 0.54 pong-2 0.55 网球-0 0.71 网球-1 0.73 0.76 网球-2 0.71 足球-0 0.66 足球-1 0.67 0.77 足球-2 0.67 rug-0 0.80 0.82 rug-1（旋转） 0.80 - rug-2（旋转） 0.78 -

Figure 12: Various c estimates obtained by our method as compared against c_m obtained using alternate potential energybased estimation, which is valid only when the object has no angular momentum.

图 12：通过我们的方法获得的各种 c估计与使用基于替代势能的估计获得的 c_m进行比较，后者仅在物体没有角动量时才有效。

All the examples are instances of collision of a single object against an infinite mass object (i.e., carpeted ground).

所有示例都是单个物体与无限质量物体（即铺有地毯的地面）碰撞的实例。

relative mass estimate

相对质量估计

Figure 13: Rotating rugby ball scene.

图 13：旋转橄榄球场景。

Estimated c using our method is $\tilde{c} = 0.78$, matching no-spin measurement 0.8 as detailed in Fig. 12.

使用我们的方法估计的 c为 $\tilde{c} = 0.78$，匹配无旋转测量 0.8，如图12所示。

5.5 Limitations and Discussion

5.5 限制和讨论

A key limitation of our method is that it only applies to rigid body collisions.

我们的方法的一个关键限制是它仅适用于刚体碰撞。

In case of deformable bodies, the physical models we employ can be unsuitable.

对于可变形体，我们采用的物理模型可能不合适。

Empirically, however, we observed that our estimates are still reasonable for near rigid collisions (see Figure 17).

然而，根据经验，我们观察到我们的估计对于近刚性碰撞仍然是合理的（见图 17）。

Our algorithm in its current form can reconstruct real-world effects under the assumption of zero friction, but including a friction model in our optimization could potentially increase the accuracy of our reconstructions.

我们当前形式的算法可以在零摩擦的假设下重建现实世界的效果，但在我们的优化中包含摩擦模型可能会提高重建的准确性。

While our current pipeline still requires user input (2 orientations per parabola segment), it is worth pointing out that the user-annotated orientations represent only a small part of the overall input data.

虽然我们当前的流程仍然需要用户输入（每个抛物线段 2 个姿态），但值得指出的是，用户标注的姿态仅代表整个输入数据的一小部分。

All the 3D positions of the objects are reconstructed fully automatically.

物体的所有 3D 位置都是全自动重建的。

The user input is typically only required to refine the orientations, which are important for angular motions, and thus help to retrieve the physical parameters.

用户输入通常只需要细化姿态，这对于角运动很重要，从而有助于检索物理参数。

Also, as we only rely on a single input video, we need enough visual information about the collision event to be visible.

此外，由于我们仅依赖于单个输入视频，因此我们需要足够的有关碰撞事件的视觉信息才能可见。

For example, if a collision is happening purely along a view direction ray, our method will most likely underestimate the objects' velocities.

例如，如果碰撞纯粹沿着视线方向射线发生，我们的方法很可能会低估物体的速度。

Likewise, objects that move very quickly might not exhibit enough change of direction from gravity.

同样，移动速度非常快的物体可能不会因重力而表现出足够的方向变化。

In such a case, our method requires more accurate centroids to reliably recover a depth estimate from gravity.

在这种情况下，我们的方法需要更准确的质心才能可靠地从重力恢复深度估计。

Lastly, it is noticeable in some reconstructions (such as Figure 14a) that the objects intersect each other at collision time.

最后，一些重建结果（例如图 14a）中，物体在碰撞时会发生**相互穿透**。

This can

出现这种情况的

Figure 14: We tested our algorithm on multiple collision sequences.

图 14：我们在多个碰撞序列上测试了我们的算法。

Here we show a selection of images of input frames and corresponding reconstructed frames.

在这里，我们展示了输入帧和相应的重建帧的图像选择。

Please refer to the supplementary video.

请参阅补充视频。

(a) Box1

(a) 方框1

(b) Box2 (d) duck-elephant (e) helmet-duck

(b) Box2 (d) 鸭子-大象 (e) 头盔-鸭子

(c) Box3

(c) 方框3

happen, as our optimization does not include any explicit collision detection or resolution terms.

原因是优化中没有显式加入碰撞检测或碰撞响应项。

While this might look like a limitation at first, we believe the accurate contacts of the other scenes actually point to a very high accuracy of the reconstructions, which was achieved without taking the exact shapes of the objects into account.

虽然一开始这看起来像是一个限制，但我们相信其他场景的准确接触实际上表明重建的精度非常高，这是在不考虑物体的确切形状的情况下实现的。

6 Conclusion

6 结论

We presented SMASH, a data-driven framework for capturing, reconstructing, and authoring collision sequences.

我们提出了SMASH，一个用于捕获、重建和创作碰撞序列的数据驱动框架。

The key idea is to regularize the problem of reconstructing collisions of object

核心思想是利用刚体物理定律，为从视频中

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input video

输入视频

BLENDER interpolated position + orientation

BLENDER 插值位置 + 姿态

SMASH reconstruction

SMASH重建

Figure 15: Direct keyframe interpolation (middle) versus our reconstructed result (bottom).

图 15：直接关键帧插值（中）与我们的重建结果（下）。

Note that the interpolated sequences completely miss the collision (middle row, third from left) as it remains oblivious to any physics considerations.

请注意，插值序列完全错过了碰撞（中行，左数第三个），因为它仍然没有考虑任何物理因素。

SMASH plugin

SMASH插件

frame #20 frame #38 frame #43

帧#20 帧#38 帧#43

Figure 16: Our SMASH plugin (top) can be used to author and synchronize collisions using the recovered parameters from our reconstruction.

图 16：我们的SMASH插件（顶部）可用于使用从重建中恢复的参数来创作和同步碰撞。

Trajectories, mass ratios and c values are read in from the optimization automatically to produce a realistic collision sequence (bottom).

自动从优化中读取轨迹、质量比和c值，以生成真实的碰撞序列（底部）。

Please refer to the supplementary video.

请参阅补充视频。

pairs from videos using laws of rigid body physics.

重建成对物体碰撞的问题引入正则化约束。

We demonstrate how to reconstruct plausible collision sequences in 3D, by observing objects in motion away from the collision instant.

我们演示了如何通过观察远离碰撞瞬间的运动物体来以 3D 形式重建合理的碰撞序列。

The method outputs physical parameters that would otherwise be very difficult to acquire, e.g., the coefficient of restitution and the mass ratio of the participating objects.

该方法输出原本很难获取的物理参数，例如恢复系数和参与物体的质量比。

The information can then be readily used to author new collision sequences.

然后可以轻松地使用该信息来创作新的碰撞序列。

While we presented a first workflow for reconstructing collision sequences, many avenues for improvements remain.

虽然我们提出了第一个重建碰撞序列的工作流程，但仍然有许多改进的途径。

Our current implementation still expects user guidance to disambiguate the objects' orientations.

我们当前的实现仍然期望用户指导来消除对象姿态的歧义。

It will be an interesting avenue for future work to iteratively apply our full pipeline to automate this step.

迭代地应用我们的完整流程来自动化此步骤将是未来工作的一个有趣的途径。

Potentially, this could also be used to approximate the shape of the object, and thus make the initial scanning step unnecessary.

潜在地，这也可以用于近似物体的形状，从而使初始扫描步骤变得不必要。

Finally, we believe our method only represents a small step towards a tighter integration of video analysis, dynamic geometry acquisition, and physics simulations.

最后，我们相信我们的方法仅代表了视频分析、动态几何采集和物理模拟更紧密集成的一小步。

There are many other physical phenomena, such as deformable objects or fluids, that could be included in order to on one hand better understand what is happening in a recorded event, and on the other hand reconstruct the underlying physics for computer animation purposes.

还可以包括许多其他物理现象，例如可变形物体或流体，一方面可以更好地理解记录事件中发生的情况，另一方面可以重建基础物理以用于计算机动画目的。

Figure 17: Although designed for rigid collisions, we observed that our method still can produce reasonable estimates for near rigid collisions as shown by the minion-rugby sequence.

图 17：虽然是针对刚性碰撞而设计的，但我们观察到我们的方法仍然可以对近刚性碰撞产生合理的估计，如小兵橄榄球序列所示。

However, the reconstruction is inaccurate around the collision time (frame #34).

然而，在碰撞时间（第 34 帧）附近的重建不准确。

Acknowledgements

致谢

We thank our reviewers for their invaluable comments.

我们感谢审稿人的宝贵意见。

We also thank Gabriel Brostow, Paul Guerrero, Martin Kilian, Oisin Mac Aodha, Moos Hueting, James Hennessey, Stephan Garbin, Chris Russell, Cl´ement Godard, Peter Hedman and Veronika Benis for their help, comments and ideas.

我们还要感谢 Gabriel Brostow、Paul Guerrero、Martin Kilian、Oisin Mac Aodha、Moos Hueting、James Hennessey、Stephan Garbin、Chris Russell、Cl´ement Godard、Peter Hedman 和 Veronika Benis 的帮助、评论和想法。

This work was partially funded by the ERC Starting Grant SmartGeometry (StG-2013-335373), the ERC Starting Grant realFlow (StG-2015-637014), UCL Impact, a Google PhD Fellowship, and gifts by Adobe.

这项工作的部分资金来自 ERC 启动资助 SmartGeometry (StG-2013-335373)、ERC 启动资助 realFlow (StG-2015-637014)、UCL Impact、Google 博士奖学金以及 Adobe 的捐赠。